

FEC/ARQ Unified Correction Symbol Model

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Contents

Abstract	4
1. System Model	5
1.1 Notation	5
1.2 Glossary	6
1.3 Components	7
1.4 The Bandwidth / Latency / Reliability Triangle	8
2. Channel Model	9
2.1 Gilbert-Elliott Two-State HMM	9
2.2 Stationary Properties	9
2.3 Burst Length Distribution	10
2.4 Concrete Examples	10
3. Recovery Fundamentals	10
3.1 The Problem: Recovering Lost Symbols	10
3.2 FEC: Proactive Recovery	11
3.3 ARQ: Reactive Recovery	12
3.4 Recovery Latency and the $P_{\text{lost}}(t)$ Model	12
4. The Taper Function	14
4.1 Definition	15
4.2 Why Match the Loss Distribution?	15
4.3 Total Correction Rate	15
4.4 The Taper Never Reaches Zero	15
4.5 Real-Time Adaptation	16
5. Unified Correction Symbol Model	16
5.1 Why Unify FEC and ARQ?	16
5.2 Correction Symbols — The Unified Concept	17
5.3 Three-Stream View: Source, Repair, Retransmit	17
5.4 Per-Slot Decision	18
6. Protocol Mechanics	19
6.1 The Retransmit Buffer	19
6.2 SACK-Extended ACK	20
6.3 Per-Symbol Delivery Outcomes	21
6.4 The Triangle in Action	22
6.5 Four-Mechanism Composition	22
7. Estimation — From Observations to Channel Parameters	23
7.1 What We Observe	23
7.2 EWMA — Fast Point Estimate	23
7.3 Beta-Binomial Posterior — Uncertainty Quantification	23
7.4 BOCD — Regime-Aware Prediction	24
7.5 GE Parameter Estimation	24
7.6 ACK Loss and Self-Healing	24
7.7 Estimation Error and Overhead	25
8. The Optimization Problem	25
8.1 Formal Statement	25
8.2 Corrected Model (canonical)	25

8.3 Burst Variance Correction	27
8.4 The Corrected Optimal Correction Rate	27
8.5 Worked Examples	27
8.6 Three-Variable Optimization	28
9. Codec Overhead Integration	31
9.1 Decoder Invocation Probability	31
9.2 Effective Codec Overhead	31
9.3 Impact on METTLE at DC	32
10. Multi-Protocol Extension	32
10.1 Per-Symbol Latency Classes	32
10.2 Interleave Before vs After FEC	32
10.3 When Shared Wins	32
10.4 Extending the Formula	32
11. Verification	33
11.1 Simulation Approach	33
11.2 Analytical Predictions to Verify	33
11.3 Boundary Cases	33
11.4 Connection to Existing Benchmarks	33
12. Congestion Control Integration	34
12.1 Why Delay-Based CC is Required	34
12.2 QUIC Datagrams Bypass Quinn's CC	34
12.3 Copa vs BBR	34
12.4 Copa's Rate Formula	35
12.5 CC + Taper: The Complete Architecture	36
12.6 ECN as Opportunistic Enhancement	36
12.7 Application Back-Pressure	36
13. Multi-Path Scheduling	37
13.1 Why FEC Beats MPTCP	37
13.2 Per-Path Model	37
13.3 Unified Symbol Stream and Interleaving	37
13.4 Correction Deficit	38
13.5 Effective Delivery Time and Bandwidth	38
13.6 Scheduler Ratio Adjustment	39
13.7 Burst Protection During Ratio Adjustment	39
13.8 Interpolated Objective Function	39
13.9 QoS Priority Cascade	40
13.10 Cross-Path Retransmit	40
Appendix A: Summary of Key Formulas	40
Appendix B: Related Work	42
ACK-Based vs NACK-Based Recovery	42
Hybrid FEC-ARQ for Lossless Streaming	42
Streaming Codes over Gilbert-Elliott Channels	42
Tail Latency Optimization with Proactive FEC	43
Joint FEC/ARQ under Gilbert-Elliott	43
Water-Filling and Optimal Resource Allocation	43
What This Model Contributes	43

Appendix C: Model Extensions from Related Work	43
C.1 Correction Symbol Preemption of Source Data [Mehrotra2010] (resolved)	44
C.2 Information Debt for Exact P_fec [RLC_GE2025]	44
C.3 Analytical P_fec Bounds [Vajha2020]	45
C.4 Multipath Diversity Gain [Zeng2021]	45
C.5 DCSW Worst-Case Taper Floor	45
C.6 Burst-Aware Channel Discount (considered refinement)	46
C.7 Summary of Extensions	47
Appendix D: Open Questions	47
Remaining Open Points (full audit)	48
Appendix E: Preliminary Poisson Model (superseded)	50
E.1 FEC Recovery Probability (preliminary — superseded by Section 8.2)	50
E.2 Solving for A* (preliminary)	50
E.3 The Optimal Correction Rate Formula (preliminary — see Section 8.4)	51
E.4 Comparison with Information-Theoretic Minimum	51
References	52
Channel Models	52
Estimation and Detection	52
FEC Codes	52
Streaming Codes Theory	52
Multipath and Scheduling	53
Hybrid FEC-ARQ	53
Streaming Code Analysis over GE	53
Tail Latency	53
Congestion Control	54
Information Theory	54
TCP SACK	54
ECN and Multicast Feedback	54
Sliding Window Channel Models	54

Abstract

Data traversing lossy multipath channels (WiFi, LTE, satellite) has specific requirements: some needs low latency (VoIP), some needs high throughput (file transfer), some tolerates partial loss (sensor telemetry). Each available path has different characteristics — bandwidth, latency, and loss rate. The goal of this model is to **optimally use each path or combination of paths to satisfy the constraints of the data sent over them**, without ad-hoc tuning knobs.

Three properties — bandwidth, tail latency, and reliability — form a triangle: fix any two, the third is determined by the channel. The protocol hint (Realtime, Bulk, etc.) selects which two to fix. The model computes the optimal correction rate, taper schedule, and path allocation from measured channel parameters — no magic offsets or manual thresholds.

The core mechanism is the **correction symbol** — a unified concept that subsumes both FEC repair symbols (proactive, flexible) and ARQ source retransmits (reactive, immediately decodable). A single taper function, shaped by the Gilbert-Elliott channel model’s burst survival function, controls correction density over time. A probabilistic per-slot decision based on loss

confidence $P_{\text{lost}}(t)$ — driven by time since send — determines whether each correction is a retransmit or a repair.

The optimal correction rate $r^* = e/(1-e) + z_{\text{delta}} \times \sqrt{e \times s2_{\text{burst}} / (W(1-e))}$ combines the information-theoretic minimum with a burst-variance- corrected tail margin. Copa congestion control is recommended over BBR for its taper-compatible rate oscillation (no FEC protection gaps from ProbeRTT).

For multipath, each path runs its own Copa, taper, and GE estimator with a unified symbol stream preserving interleaved burst protection. A global correction deficit tracks outstanding recovery needs across paths. The scheduler adjusts per-path source/correction ratios using an interpolated latency/bandwidth objective, with QoS priority cascading for mixed traffic.

ACK-based feedback with SACK replaces the NACK mechanism — the same proven approach TCP has used for 40 years, with sender-side loss inference from ACK absence.

1. System Model

1.1 Notation

Symbol	Meaning	Unit	Example
ϵ	Average channel loss rate	probability (0-1)	0.025 (2.5% WiFi)
p	$P(\text{Good} \rightarrow \text{Bad})$ in GE model	probability (0-1)	0.013
q	$P(\text{Bad} \rightarrow \text{Good})$ in GE model	probability (0-1)	0.5
B	Mean burst length = $1/q$	symbols (count)	2.0
W	Encoder window size	symbols (count)	50
RTT	Round-trip time	seconds	0.050 (50ms)
$P_{\text{lost}}(t)$	$P(\text{symbol lost given no ACK after time } t)$	probability (0-1)	0.5 at $t=\text{SRTT}$
t_{fec}	FEC recovery time = $m / (A \times (1-\epsilon)) \times t_{\text{sym}}$	seconds	0.003 (3ms)
t_{sym}	Symbol transmission time = $\text{symbol_size} / \text{throughput}$	seconds	0.000096 (96 μ s at 100Mbps)
SRTT	Smoothed RTT estimate	seconds	0.050 (50ms)
RTTVAR	RTT variance estimate (standard deviation)	seconds	0.005 (5ms)
r	Total correction rate	ratio (corrections/source)	0.08 (8%)
$\tau(t)$	Taper function: correction density at offset t	ratio (corrections/symbol)	0.04
A	Taper amplitude (scaling factor)	ratio (corrections/symbol)	0.04
P_{fec}	Probability a lost symbol is FEC-recovered	probability (0-1)	0.95
δ	Tail latency target: $P(\text{late delivery}) \leq \delta$	probability (0-1)	1e-4

Symbol	Meaning	Unit	Example
ρ	Reliability target: $P(\text{symbol delivered}) \geq \rho$	probability (0-1)	1.0 (100%)
T_{cut}	Taper cutoff time (stop corrections after this)	seconds	∞ (100% reliability)
B_{max}	99.99th percentile burst length = $\text{ceil}(\ln(0.0001)/\ln(1-q))$	symbols (count)	19 (WiFi, $q=0.5$)
buffer_max	Max retransmit buffer size (derived)	symbols (count)	750 (WiFi 100Mbps)
L_{prop}	Propagation delay (base latency)	seconds	0.025 (25ms)
L_{arq}	ARQ recovery latency (time from loss to retransmit arrival)	seconds	0.075-0.100
σ^2_{burst}	Burst variance inflation factor	dimensionless	2.9
z_{δ}	Standard normal quantile for δ	dimensionless	3.72 (for $\delta=1e-4$)
$\varepsilon_{\text{burst}}$	Current channel loss rate (fast EWMA or GE state, optional — see C.6)	probability (0-1)	0.5 (during burst)
$\varepsilon_{\text{codec}}$	Codec decode overhead	ratio (0-1)	0.01 (RaptorQ)

1.2 Glossary

Correction symbol — any symbol sent to recover lost data. Either a source retransmit (exact copy, immediate decode) or a repair symbol (FEC linear combination, needs decoder). The taper function controls how many; $P_{\text{lost}}(t)$ controls which type. See Section 5.2.

Repair symbol — a correction symbol generated by the FEC encoder. It is a random linear combination of source symbols in the encoder window over GF(256). The receiver feeds it to the decoder. See Section 3.2.

Source retransmit — a correction symbol that is an exact copy of a previously-sent source symbol from the retransmit buffer. The receiver can use it immediately without decoding. See Section 3.3.

Encoder window — the W most recent source symbols tracked by the FEC encoder. Repair symbols are linear combinations of symbols in this window. When a symbol is evicted (window slides forward), it can no longer be covered by new repair symbols. See Section 3.2.

Taper function $\tau(t)$ — specifies the correction symbol density at time offset t from a source symbol. Shaped to match the channel's burst-length distribution: $\tau(t) = A \times (1-q)^t$ for Gilbert-Elliott channels. See Section 4.

Systematic code — an FEC scheme where source symbols are sent as-is (not encoded). The decoder is only invoked when at least one source symbol is lost. If nothing is lost, the data passes through without any decoding cost.

Codec overhead — the extra symbols the decoder needs beyond the number of losses, because not all random linear combinations are linearly independent. E.g., RaptorQ needs ~1% extra, METTLE needs ~15%. See Section 9.

SACK (Selective Acknowledgement) — an extension to cumulative ACK that reports out-of-order received blocks beyond the cumulative point. Lets the sender know exactly which symbols arrived despite gaps. See Section 6.1.

GE states (Good/Bad) — the two states of the Gilbert-Elliott channel model. In the simplified model: Good = no loss, Bad = total loss. Transition probabilities p (Good→Bad) and q (Bad→Good) determine burst behavior. See Section 2.

P_{lost}(t) — probability that a specific symbol was lost, given no ACK received after time t . The per-slot mixing probability for repair vs retransmit. See Section 3.4.

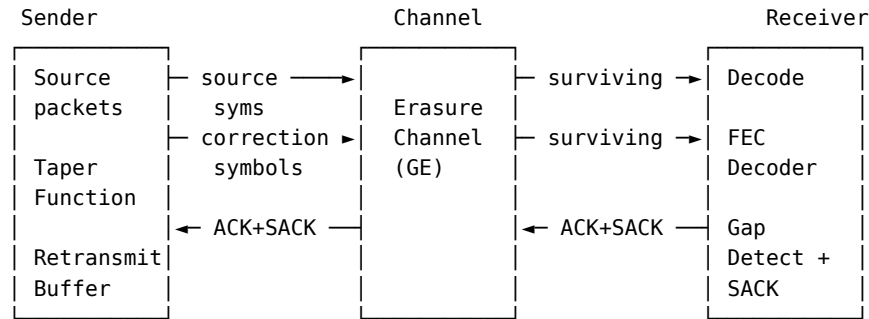
P_{fec} — probability that a lost symbol is recovered by FEC (proactive correction) before ARQ retransmit kicks in. See Section 8.2.

P_{arq} — probability that ARQ retransmit succeeds, given FEC failed. Derived from reliability target: $P_{arq} = 1 - (1 - p) / (\epsilon \times (1 - P_{fec}))$. See Section 6.3.

z_{delta} — standard normal quantile at probability $(1 - \delta)$. Controls the tail margin in the repair rate formula. See Section 8.4.

σ^2 _{burst} (s2_{burst}) — burst variance inflation factor for the GE channel: $1 + 2(1 - p - q) / (p + q)$. Corrects the iid normal approximation for burst-correlated losses. See Section 8.3.

1.3 Components



The system provides **reliable** delivery. Three properties form a triangle — bandwidth, tail latency, and reliability. Fix any two, the third is determined by the channel. The protocol hint selects which two to fix.

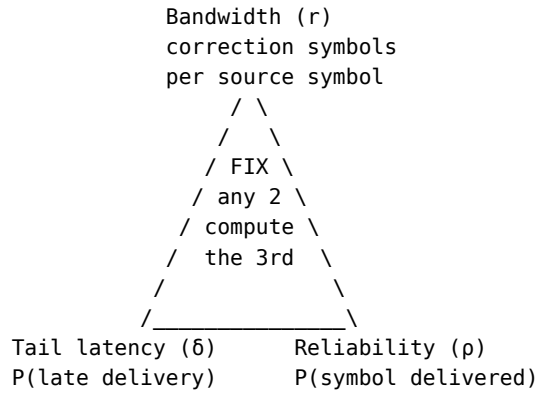
One unified mechanism — **correction symbols** — handles both proactive and reactive recovery. The taper function controls correction symbol density, and each correction slot either retransmits an exact source symbol from the retransmit buffer (preferred, if old enough) or generates a random repair symbol (FEC fallback):

Aspect	Correction Symbol (retransmit)	Correction Symbol (FEC repair)
When chosen	With probability $P_{lost}(t)$	With probability $1 - P_{lost}(t)$
Content	Exact copy of source symbol	Random linear combination
Decode cost	Zero (immediate use)	Needs FEC decoder

Aspect	Correction Symbol (retransmit)	Correction Symbol (FEC repair)
Bandwidth cost	Same (one symbol slot)	Same (one symbol slot)
Latency cost	Depends on when P_lost triggers	Zero additional (arrives with source)

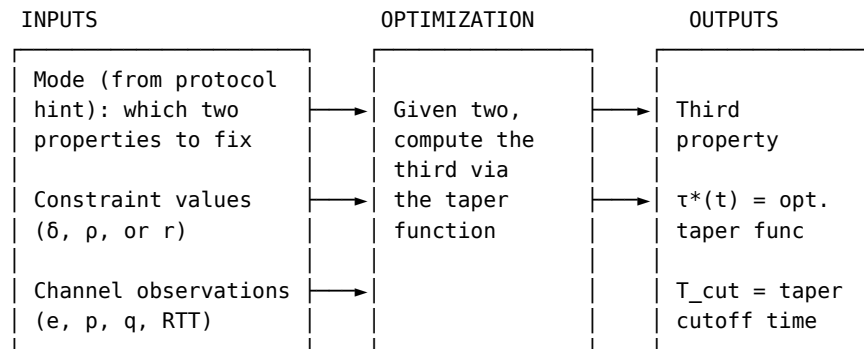
1.4 The Bandwidth / Latency / Reliability Triangle

Three properties are linked by the channel. Fix any two, the third is determined:



Mode	Fix	Compute	Use case
Bulk transfer	$\rho=100\%$, minimize r	δ (tail latency)	File transfer, backup
VoIP	δ (max latency), r (codec rate)	ρ (reliability)	Interactive voice
Live video	δ (frame deadline), ρ ($\geq 99.9\%$)	r (bandwidth)	Streaming, conferencing
Gaming	δ (tight), ρ ($\geq 99\%$)	r (bandwidth)	Real-time game state
Sensor/IoT	r (minimal), ρ ($\geq 95\%$)	δ (tail latency)	Periodic telemetry

The protocol hint selects the mode and constraints:



When $\rho < 100\%$, the taper has a finite cutoff T_{cut} . Symbols not recovered by T_{cut} are permanently lost. When $\rho = 100\%$, $T_{\text{cut}} = \infty$ (correction symbols continue until ACK, the infinite-tailed taper from Section 4).

$$\pi_B = p / (p + q) \quad \text{probability of being in Bad state}$$

$$\pi_G = q / (p + q) \quad \text{probability of being in Good state}$$

Average loss rate:

$$e = \pi_B \times h_B + \pi_G \times h_G = \pi_B = p / (p + q)$$

2.3 Burst Length Distribution

Given we just entered the Bad state, the burst length T follows a geometric distribution:

$$P(T = t) = q \times (1-q)^{(t-1)} \quad \text{for } t = 1, 2, 3, \dots$$

$$P(T \geq t) = (1-q)^{(t-1)} \quad \text{survival function}$$

$$E[T] = 1/q = B \quad \text{mean burst length}$$

This survival function is the key quantity — it tells us, given that a burst started, how likely it is to still be ongoing after t symbols.

2.4 Concrete Examples

Scenario	ϵ (loss)	p ($G \rightarrow B$)	q ($B \rightarrow G$)	$B = 1/q$	Character
DC	0.1%	0.001	0.5	2.0	Rare, short bursts
WiFi	2.5%	0.013	0.5	2.0	Moderate, short
LTE	5%	0.02	0.4	2.5	Moderate, medium
Satellite	9%	0.03	0.3	3.3	Frequent, long
Bad WiFi	15%	0.05	0.3	3.3	Frequent, long

3. Recovery Fundamentals

3.1 The Problem: Recovering Lost Symbols

When a source symbol is erased by the channel, the receiver has a gap in the data stream. There are two fundamentally different ways to fill that gap:

1. **Send extra data proactively** (before knowing what will be lost): the sender generates redundant symbols alongside source data. If a source symbol is lost, the redundant symbols can reconstruct it. This costs bandwidth (the redundant symbols are sent whether needed or not) but adds no latency (they arrive at roughly the same time as the source).
2. **Detect the loss and resend** (after knowing what was lost): the sender discovers that a symbol wasn't received (via ACK timeout) and retransmits it. This costs latency (at least one round-trip to detect + retransmit) but wastes no bandwidth (only lost symbols are resent).

These two approaches are called **FEC** (Forward Error Correction) and **ARQ** (Automatic Repeat reQuest). The challenge is finding the right balance between them — more FEC means lower latency but higher bandwidth; more ARQ means lower bandwidth but higher latency.

3.2 FEC: Proactive Recovery

The encoder window. The sender maintains a sliding window of the W most recent source symbols. This window is the encoder's "memory" — it tracks which source symbols are available for generating repair:

Encoder window ($W = 5$ symbols):

```
sent:  [S1] [S2] [S3] [S4] [S5] [S6] [S7] [S8] ...
           |----- window -----|
           S3  S4  S5  S6  S7
```

As new symbols arrive, old ones are evicted from the left.

Repair symbols are random linear combinations of source symbols in the window. Each repair symbol is computed as:

$$R = c1*S3 + c2*S4 + c3*S5 + c4*S6 + c5*S7$$

where $c1..c5$ are random coefficients in $GF(256)$ (a finite field — arithmetic wraps around so values stay within 0-255). Each repair symbol is a different random combination, giving the decoder independent equations.

How decoding works. If k source symbols in the window are lost, the receiver needs at least k repair symbols that survived the channel. Each repair symbol provides one linear equation relating the lost symbols. With k equations and k unknowns, the decoder solves the system (Gaussian elimination over $GF(256)$) to recover the lost data.

Example: source S3 and S5 lost, S4/S6/S7 received

Time ----->

```
Source:  [S1] [S2] [S3] [S4] [S5] [S6] [S7]
           X           X           <- lost
Repair:  [R1] [R2] [R3] [R4]
           |   |   |   |
           v   v   v   v
        covers covers covers covers
        S1-S3  S2-S4  S3-S6  S5-S7
```

Decoder receives R1, R2, R3, R4 and knows S4, S6, S7.

Substituting known values into R3 and R4:

$$R3 = c1*S3 + c2*(\text{known } S4) + c3*S5 + c4*(\text{known } S6)$$

$$R4 = c5*S5 + c6*(\text{known } S6) + c7*(\text{known } S7)$$

-> 2 equations, 2 unknowns (S3, S5) -> solve.

Systematic codes. In our system, source symbols are sent as-is — they are not encoded. This means if nothing is lost, the receiver can use the data immediately without any decoding. The decoder is only invoked when at least one source symbol is missing. This is important for efficiency: at low loss rates, the decoder rarely runs.

Codec overhead. Not all random linear combinations are independent. Some may be (near-)linearly dependent, providing redundant equations. This means the decoder sometimes needs slightly more repair symbols than the number of losses. This extra requirement is the codec overhead — e.g., 1% for RaptorQ means the decoder needs 1% more symbols than the theoretical minimum of k .

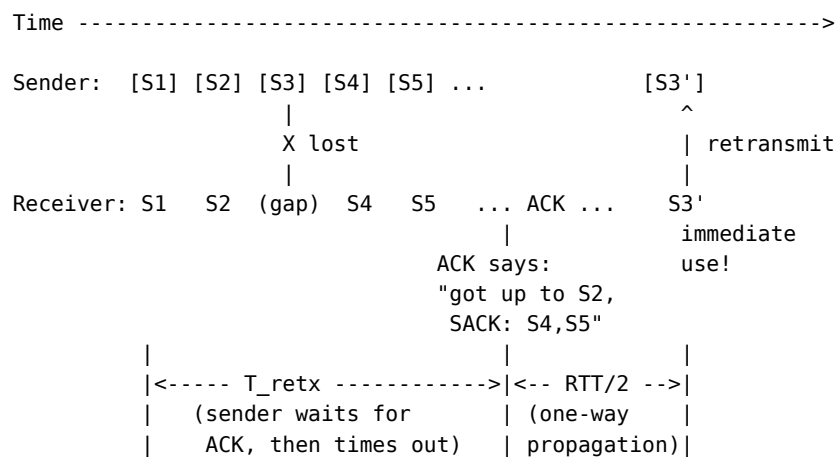
Properties of FEC: - Bandwidth cost: r repair symbols per source symbol (always-on, paid whether loss occurs or not) - Latency cost: zero additional (repair arrives at roughly the same time as source) - Bandwidth fraction used: $r/(1+r)$ of total link capacity

3.3 ARQ: Reactive Recovery

ARQ recovers lost symbols by retransmitting them after detecting the loss. In our system, loss detection is **sender-side**: the sender tracks which symbols have been ACKed and retransmits any that remain un-ACKed after a timeout.

This is how TCP has worked for 40 years [RFC2018] and is proven robust for unicast connections. The alternative (receiver-side detection via NACK messages) adds detection delay and is fragile when the reverse path is lossy.

Timeline of ARQ recovery:



The sender sent S3 at time 0. After T_{retx} (roughly one RTT, enough time for an ACK to return), no ACK has confirmed S3. The sender retransmits S3. The retransmitted symbol travels one-way ($RTT/2$) to the receiver.

Key advantage over FEC: A retransmitted source symbol is immediately usable by the receiver — no decoder needed, no dependency on other symbols. The receiver gets the exact data it was missing and can process it right away.

Key disadvantage: The retransmission adds $L_{\text{arq}} = T_{\text{retx}} + RTT/2$ of latency. For a link with 50ms RTT and $T_{\text{retx}} = 75\text{ms}$: $L_{\text{arq}} = 100\text{ms}$. This is acceptable for bulk transfer but too slow for real-time applications.

Properties of ARQ: - Bandwidth cost: approximately ϵ per source symbol (only retransmit what's actually lost — no waste) - Latency cost: $L_{\text{arq}} = T_{\text{retx}} + RTT/2$ per recovery event - Self-healing: if the ACK itself is lost on the reverse path, the sender just waits longer and retransmits anyway. No special mechanism needed.

3.4 Recovery Latency and the $P_{\text{lost}}(t)$ Model

A lost symbol is recovered by whichever mechanism finishes first — FEC or ARQ. Rather than a hard timeout threshold, the choice between repair and retransmit is **probabilistic**, based on the sender's confidence that a symbol was lost.

FEC recovery time (t_{fec}): The taper function generates repair symbols continuously after a source symbol is sent. Each repair that covers the lost symbol AND survives the channel gives the decoder one more equation. For m lost symbols in the window, the decoder needs m surviving repairs:

$$t_{\text{fec}} = m / (A \times (1-e)) \times t_{\text{sym}}$$

where:

m = number of lost symbols in the window (usually 1)
 A = taper amplitude (corrections/symbol at offset 0)
 $1-e$ = probability each repair survives the channel
 t_{sym} = symbol_size / throughput (time to transmit one symbol)

Concrete examples for a single loss ($m=1$), $A=0.04$, $e=0.025$:

At 100 Mbps, 1200-byte symbols: $t_{\text{sym}} = 0.096\text{ms}$, $t_{\text{fec}} = 2.5\text{ms}$
 At 10 Mbps, 1200-byte symbols: $t_{\text{sym}} = 0.96\text{ms}$, $t_{\text{fec}} = 24.6\text{ms}$
 At 1 Mbps, 1200-byte symbols: $t_{\text{sym}} = 9.6\text{ms}$, $t_{\text{fec}} = 245\text{ms}$

For burst loss ($m=5$ on WiFi at 100 Mbps): $t_{\text{fec}} = 5 \times 2.5\text{ms} = 12.5\text{ms}$.

$P_{\text{lost}}(t)$: the probability a symbol was lost. At time t after sending a symbol, given no ACK has arrived, what is the probability it was actually lost?

If the symbol was received, the ACK should arrive after roughly one RTT. If the symbol was lost, no ACK will ever come (until a correction recovers it). Using Bayes' theorem:

$$P_{\text{lost}}(t) = e / [e + (1-e) \times P(\text{RTT} > t)]$$

where:

e = channel loss rate (prior probability of loss)
 $P(\text{RTT} > t)$ = probability the ACK is delayed beyond time t
 = tail of the RTT distribution (from SRTT and RTTVAR)

Assuming RTT is normally distributed with mean SRTT and standard deviation RTTVAR: $P(\text{RTT} > t) = 1 - \Phi((t - \text{SRTT}) / \text{RTTVAR})$, where Φ is the standard normal CDF. This is the same normal approximation used in TCP's RTO computation [RFC6298].

This gives a smooth transition from “probably fine” to “certainly lost”:

$t = 0$:	$P_{\text{lost}} = e$	(just the base loss rate)
$t = \text{SRTT}$:	$P_{\text{lost}} \approx 2e$	(ACK expected by now)
$t = \text{SRTT} + 2s$:	$P_{\text{lost}} \approx 0.98$	(very confident it's lost)
$t \gg \text{SRTT}$:	$P_{\text{lost}} \rightarrow 1.0$	(certainly lost)

Concrete example (WiFi, $e = 0.025$, SRTT = 50ms):

$t = 0\text{ms}$:	$P_{\text{lost}} = 0.025$	-> 97.5% repair, 2.5% retransmit
$t = 40\text{ms}$:	$P_{\text{lost}} = 0.08$	-> 92% repair, 8% retransmit
$t = 50\text{ms}$:	$P_{\text{lost}} = 0.05$	-> 95% repair, 5% retransmit
$t = 60\text{ms}$:	$P_{\text{lost}} = 0.35$	-> 65% repair, 35% retransmit
$t = 70\text{ms}$:	$P_{\text{lost}} = 0.85$	-> 15% repair, 85% retransmit
$t = 80\text{ms}$:	$P_{\text{lost}} = 0.98$	-> 2% repair, 98% retransmit

The per-slot decision uses $P_{\text{lost}}(t)$ directly:

$$P(\text{retransmit}) = P_{\text{lost}}(t_k)$$

No hard threshold. The transition from repair to retransmit is driven by TIME since send. Early (t small): P_{lost} is low, so corrections are mostly repair — flexible coverage for uncertain

losses. Late ($t \gg \text{SRTT}$): P_{lost} approaches 1, so corrections are mostly retransmit — targeted recovery of confirmed losses. No tuning knobs.

See Appendix C.6 for an optional channel-state discount (ϵ_{burst}) that provides modest improvement for long-burst scenarios.

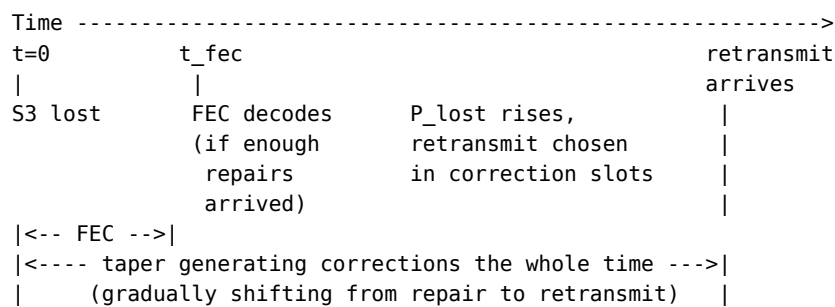
Bandwidth efficiency. A retransmit is wasted if the symbol wasn't actually lost (duplicate). A repair is never wasted (always provides information to the decoder). The expected waste per correction slot is:

$$\begin{aligned} E[\text{waste}] &= P(\text{retransmit chosen}) \times P(\text{symbol not actually lost}) \\ &= P_{\text{lost}}(t) \times (1 - P_{\text{lost}}(t)) \end{aligned}$$

This is maximized at $P_{\text{lost}} = 0.5$ (maximum uncertainty) and zero at the extremes. The probabilistic model automatically minimizes waste.

Proactive retransmit emerges naturally. At high loss rates (e.g., $e = 0.5$), $P_{\text{lost}}(0) = 0.5$ — half the correction slots are retransmits even at $t = 0$. This is proactive redundant source sending, no special mode needed. At low loss rates ($e = 0.001$), $P_{\text{lost}}(0) = 0.001$ — virtually all repair. The model adapts to the channel automatically.

Which mechanism wins on latency? FEC recovery (t_{fec}) is typically much faster than ARQ recovery (waiting for P_{lost} to rise + $\text{RTT}/2$):



At 100 Mbps on WiFi: $t_{\text{fec}} = 2.5\text{ms}$. Most losses are FEC-recovered long before P_{lost} rises high enough for retransmission. ARQ retransmit is only relevant for burst losses that overwhelm the FEC budget.

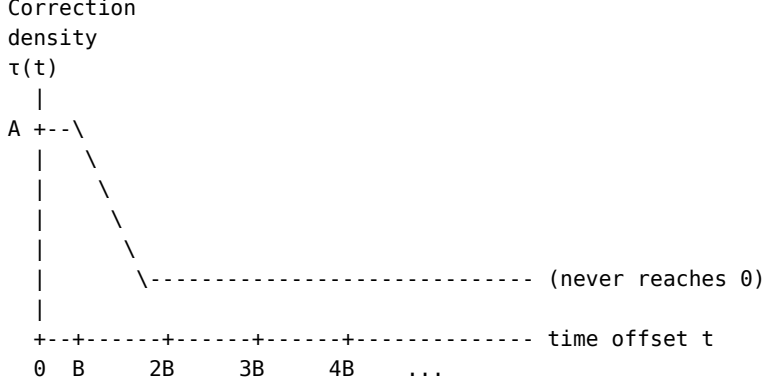
4. The Taper Function

Why not send correction symbols at a constant rate? If loss were uniformly distributed over time (i.i.d.), a constant rate would be optimal — every position is equally likely to need correction. But on real wireless channels, loss comes in **bursts** (Section 2). A burst wipes out consecutive symbols, then the channel recovers. Right after a burst starts, the probability of continued loss is high; as time passes, it decays exponentially.

A constant correction rate misallocates: it sends too many corrections during good periods (wasted) and too few right after a burst (when they're needed most). The taper function solves this by matching correction density to the burst survival probability — more corrections where loss is likely, fewer where it isn't. This minimizes the total correction budget for a given recovery target.

4.1 Definition

The taper function $\tau(t)$ specifies the correction density at time offset t from a source symbol. At offset t after symbol s enters the window, we generate $\tau(t)$ correction symbols covering s .



$$\tau(t) = A \times (1-q)^t$$

A = amplitude (what we solve for)
 $(1-q)^t$ = GE burst survival function

4.2 Why Match the Loss Distribution?

The taper should allocate more correction where loss is more likely. Given that a symbol was lost (we're in a burst), the conditional probability that the burst is still active at offset t is:

$$P(\text{burst active at offset } t \mid \text{burst at offset } 0) = (1-q)^t$$

The optimal correction allocation is proportional to this conditional probability. This is the **water-filling solution**: given a fixed budget, allocate resources proportional to the probability of needing them.

Proof sketch (Lagrange multipliers): We want to maximize P_{fec} given a fixed total correction budget r . The marginal benefit of a correction symbol at offset t is proportional to $P(\text{burst still active at } t)$. The Lagrangian is maximized when the correction density is proportional to $(1-q)^t$. This water-filling principle is analogous to the delay-optimal streaming code constructions in [Badr2017] and [Fong2019].

For an i.i.d. channel ($q = 1$, no burst memory): $\tau(t) = \text{constant}$ (flat taper). This is correct — every position is equally likely to need correction.

4.3 Total Correction Rate

The total correction rate (correction symbols per source symbol) is:

$$r = \sum_{t=0}^{\infty} \tau(t) = A \times \sum_{t=0}^{\infty} (1-q)^t = A / q$$

Since $0 < q \leq 1$, this geometric series converges. Therefore:

$$A = r \times q$$

The amplitude is uniquely determined by the correction rate and the GE parameter.

4.4 The Taper Never Reaches Zero

The exponential $(1-q)^t$ is always positive for $0 < q < 1$. This is correct behavior: there is always a nonzero probability of a burst still continuing. As long as a symbol has not been ACK'd,

there is a nonzero probability it was lost, so we should continue generating (increasingly rare) correction coverage.

$t = 0:$	$\tau(0) = A$	peak correction density
$t = B:$	$\tau(B) = A \times e^{-1} \approx 0.37 \times A$	one mean burst length
$t = 2B:$	$\tau(2B) = A \times e^{-2} \approx 0.14 \times A$	two mean burst lengths
$t = 5B:$	$\tau(5B) = A \times e^{-5} \approx 0.007 \times A$	five mean burst lengths
$t \rightarrow \infty:$	$\tau(t) \rightarrow 0$	but never zero

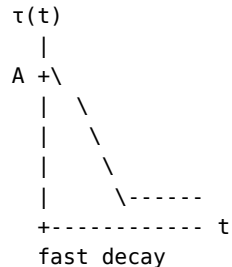
In practice, once a symbol is ACK'd, we stop generating correction for it (the encoder window advances past it). The theoretical infinite tail is truncated by the ACK mechanism.

4.5 Real-Time Adaptation

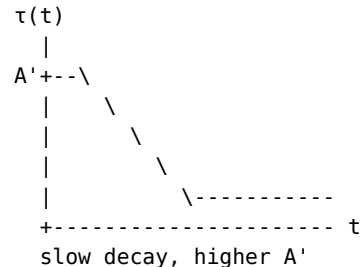
The taper function adapts in real time through two mechanisms:

1. **GE parameter updates:** The estimator continuously tracks q (and p) from observed loss patterns. As q changes, the taper shape changes — slower decay for longer bursts, faster decay for shorter bursts.
2. **BOCD changepoint detection:** If the loss regime changes abruptly (e.g., path switches from WiFi to LTE), BOCD detects the changepoint within 5-15 batches (a batch = one ACK feedback cycle, typically at intervals of 10-100ms depending on sending rate and path RTT) and widens the posterior, increasing the correction budget until the new regime is characterized.

Before changepoint:
(short bursts, $q=0.5$)



After changepoint:
(long bursts, $q=0.2$)



5. Unified Correction Symbol Model

5.1 Why Unify FEC and ARQ?

Both FEC and ARQ produce one symbol on the wire. Both cost the same bandwidth per symbol. The difference is timing and content:

FEC repair:	generated immediately, random combination, needs decoder
ARQ retransmit:	generated after timeout, exact source copy, immediately usable

Both occupy one symbol slot on the wire.
Both are subject to the same channel loss rate e .
Both serve the same purpose: recover a lost source symbol.

The key insight: **the taper function controls WHEN to generate correction symbols** (the density schedule). **The per-slot decision controls WHAT to generate** (repair or retransmit). These are orthogonal choices:

- **Early in the taper** (right after sending source): $P_{\text{lost}}(t)$ is low, so almost all correction symbols are FEC repair. This is pure proactive protection — we don't yet know what's lost.
- **Late in the taper** ($t \gg \text{SRTT}$): $P_{\text{lost}}(t)$ approaches 1, so correction symbols are almost all retransmits. We're confident about which symbols are lost and send the exact data the receiver needs.
- **In between**: a smooth probabilistic mix. No hard phase switch.

This means the taper function **naturally transitions from FEC to ARQ** as time passes, driven by the $P_{\text{lost}}(t)$ posterior. The unified mechanism is called a **correction symbol** — it is either a repair symbol or a source retransmit, decided probabilistically at generation time.

5.2 Correction Symbols — The Unified Concept

A **correction symbol** is any symbol sent to recover lost data. It occupies one symbol slot on the wire and serves one of two purposes:

Aspect	Source retransmit (ARQ)	Repair symbol (FEC)
When chosen	With probability $P_{\text{lost}}(t)$	With probability $1 - P_{\text{lost}}(t)$
Content	Exact copy of source symbol	Random linear combination
Receiver action	Immediate use, no decoder	Feed to FEC decoder
Bandwidth cost	Same (one symbol slot)	Same (one symbol slot)
Waste risk	Duplicate if symbol wasn't lost	Never wasted (always useful)
Best for	Confirmed loss + good channel	Uncertain loss or burst (flexible)

The taper function (Section 4) determines the **density** of correction symbols over time. This section explains what happens in each slot.

5.3 Three-Stream View: Source, Repair, Retransmit

Conceptually, the sender manages **three streams** that compete for wire capacity. Each stream has a different effect on the bandwidth/latency/ reliability triangle:

Stream	Latency	Bandwidth	Reliability
Source	++ immediate	neutral	neutral
Repair (FEC)	- decoder wait	+ (no waste if taper matched)	++ (covers any loss)
Retransmit (ARQ)	+ immediate decode	- (duplicate risk)	+ (targeted recovery)

The hierarchical model (taper + P_{lost}) determines the three-stream mix:

Total capacity: C (from Copa)

source_rate	= $C / (1 + r)$	[from taper ratio r]
retransmit_rate	= $C \times r / (1+r) \times P_{\text{lost}}(t)$	[confirmed losses]
repair_rate	= $C \times r / (1+r) \times (1 - P_{\text{lost}}(t))$	[uncertain losses]

where r = taper correction ratio (Section 4)
 $P_{\text{lost}}(t)$ = loss confidence (Section 3.4)

The three rates sum to C. The taper ratio r controls the source/correction split. $P_{\text{lost}}(t)$ controls the repair/retransmit split within corrections.

How the protocol hint shifts the mix:

Realtime (low latency):

- > tight delta -> high r (more corrections for fast FEC recovery)
- > BUT also high P_{retx} when confident (retransmit = immediate decode)
- > Mix: moderate source, moderate repair, moderate retransmit

Bulk (high bandwidth):

- > loose delta -> low r (fewer corrections, more source)
- > low P_{retx} (prefer repair over retransmit, less duplicate risk)
- > Mix: lots of source, some repair, little retransmit

VoIP (fixed bandwidth + latency, variable reliability):

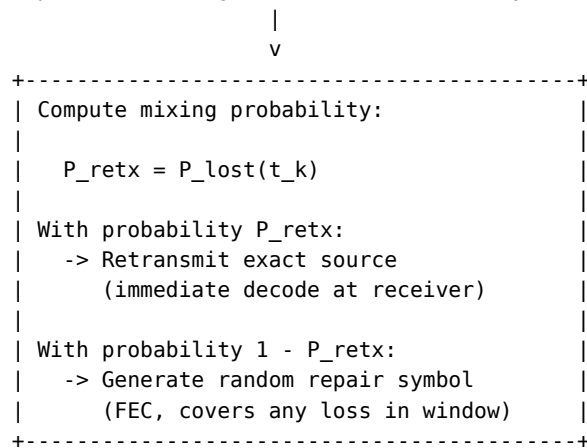
- > r constrained by bandwidth budget
- > high P_{retx} when confident (immediate decode matters)
- > Mix: fixed source rate, split corrections by P_{retx}

Per-path three-stream fractions: Each path has its own r_i and P_{retx_i} , so the three-stream fractions naturally differ per path. A low-latency path with low loss has: high source, low repair, low retransmit. A high-loss path has: lower source, high repair, moderate retransmit. The hierarchical model produces these automatically — no separate three-way optimization needed.

5.4 Per-Slot Decision

When the taper function (Section 4) decides to generate a correction symbol, the sender makes a probabilistic choice based on $P_{\text{lost}}(t)$ for the oldest un-ACKed symbol in the retransmit buffer:

Taper decides: "generate a correction symbol now"



$$P_{\text{lost}}(t_k) = e / [e + (1-e) \times P(\text{RTT} > t_k)] \quad \text{per-symbol loss confidence}$$

The FEC/ARQ mix is regulated by TIME since send. Early correction slots (t small) have low P_{lost} — the ACK hasn't had time to return, so loss is uncertain. Almost all corrections are FEC repair, which flexibly covers any lost symbol. Late correction slots ($t \gg \text{SRTT}$) have high P_{lost} — the ACK should have arrived by now, so loss is confirmed. Corrections shift to retransmit, which is immediately decodable.

This naturally produces Mehrotra & Li’s optimal policy [Mehrotra2010]: during bursts (before RTT elapses), P_{lost} stays low $\rightarrow$ mostly repair. After bursts (ACKs reveal gaps), P_{lost} rises $\rightarrow$ targeted retransmit.

P_{retx} is zero at the extremes ($P_{\text{lost}} = 0$: all repair, no waste; $P_{\text{lost}} = 1$: all retransmit, definitely lost so no waste) and maximized at $P_{\text{lost}} = 0.5$ (maximum uncertainty). The model automatically allocates the right mix.

See Appendix C.6 for an optional channel-state discount (ϵ_{burst}) that provides modest improvement for long-burst scenarios.

6. Protocol Mechanics

6.1 The Retransmit Buffer

The sender maintains a **retransmit buffer**: an ordered list of source symbols that have been sent but not yet ACKed.

Retransmit buffer (ordered by send time, oldest first):

```
+-----+-----+-----+-----+-----+-----+-----+
| S12 | S13 | S17 | S18 | S19 | S24 | S25 | <- un-ACKed symbols
+-----+-----+-----+-----+-----+-----+-----+
  ^                                     ^
oldest                               newest
(first retransmit                     (too recent,
 candidate if                          wait for ACK)
 age > T_retx)
```

Operations: - **Enqueue**: every sent source symbol is added to the tail - **Dequeue (ACK)**: when an ACK or SACK confirms receipt, remove the symbol - **Peek (retransmit)**: when generating a correction symbol, check the head — if its age exceeds T_{retx} , retransmit it (but keep it in the buffer until ACKed)

The retransmit buffer is NOT bounded by the encoder window. The encoder window (W symbols) bounds what FEC repair can cover. The retransmit buffer bounds what ARQ retransmit can cover. They serve different purposes:

- Symbol in encoder window AND buffer: protected by FEC + ARQ
- Symbol evicted from window, still in buffer: protected by ARQ only
- Symbol ACKed: removed from buffer (delivered)

Two independent mechanisms manage the buffer:

Age eviction (T_{cut}): Symbols older than T_{cut} (derived from the reliability target ρ in Section 8.6) are evicted from the buffer — accepted as permanently lost. T_{cut} is derived from the triangle optimization (Section 8.6): for reliability target ρ , T_{cut} is the time where $P(\text{not recovered}) = 1 - \rho$. When $\rho = 100\%$, the equation has no finite solution, so $T_{\text{cut}} = \infty$ (never evict). This emerges from the math — not a special case.

Size backpressure (buffer_max): When the buffer reaches `buffer_max` entries, the sender pauses accepting new source symbols from the application (backpressure). Correction symbols continue being generated for existing buffer contents — corrections recover lost symbols, ACKs arrive, and the buffer drains. Once space is available, source resumes.

buffer_max is derived, not configured:

For $\rho < 100\%$: $\text{buffer_max} = \text{source_rate} \times T_{\text{cut}}$
(eviction keeps buffer within this bound)

For $\rho = 100\%$: $\text{buffer_max} = \text{source_rate} \times (\text{RTT} + B_{\text{max}} / (r^* \times (1-e)) \times t_{\text{sym}})$
where $B_{\text{max}} = \text{ceil}(\ln(0.0001) / \ln(1-q)) \approx 9.2/q$
(99.99th percentile burst length from GE model)

Both mechanisms always run. T_{cut} determines which triggers first: - Finite $T_{\text{cut}} \rightarrow$ eviction keeps buffer small $\rightarrow$ backpressure rarely triggers - $T_{\text{cut}} = \infty \rightarrow$ no eviction $\rightarrow$ backpressure is the only limit

What bends when: - $\rho < 100\%$: reliability bends (eviction at T_{cut}) - $\rho = 100\%$: latency bends (backpressure at buffer_max, rare)

6.2 SACK-Extended ACK

The receiver sends an **ACK per received symbol** — every incoming symbol triggers an immediate ACK in the reverse direction. This provides the fastest possible loss detection and the most RTT samples for estimation.

Overhead: Each ACK is ~80 bytes (fields + QUIC/UDP headers). At any link speed, ACK traffic is approximately 0.6% of the data rate — negligible.

Alternatives considered: - Per-batch ACK: lower overhead, but slower loss detection (batch granularity) - Delayed ACK (TCP style, every 200ms): 50% less traffic, but adds 200ms detection delay — too slow for our P_{lost} model - Piggyback ACK on reverse-direction data: zero overhead for bidirectional traffic — a future optimization for the bidirectional case

ACK contents (5 fields):

```
+-----+
| cumulative_ack:      u64    all seqs <= this received |
| sack_ranges:        [(u64,u64)] additional ranges   |
| echo_timestamp:      u64    sender's timestamp      |
| jitter_us:           u32    interarrival jitter (us) |
| cumulative_received: u64    total symbols received  |
+-----+
```

- **cumulative_ack**: highest sequence number such that ALL symbols up to and including it have been received. An optimization — equivalent to a SACK range starting at 0, but compressed to one number.
- **sack_ranges** (Selective ACK [RFC2018]): out-of-order ranges received beyond the cumulative point. Tells the sender exactly which symbols arrived despite gaps. Cumulative within the T_{cut} window — all received fragments are reported, not just the most recent (unlike TCP which limits to 3-4 blocks).
- **echo_timestamp**: sender's own send timestamp echoed back for RTT measurement. $\text{RTT} = \text{now} - \text{echo}$. No clock synchronization needed.
- **jitter_us**: interarrival jitter in microseconds [RFC3550 A.8]. u32 holds up to 4300 seconds (matches RFC 3550's 32-bit field). Used for reorder buffer sizing, congestion detection, and real-time jitter budgets.
- **cumulative_received**: running total of symbols received. Self-healing counter that survives ACK loss and SACK pruning. Gives the sender a reliable aggregate reliability metric: $\rho_{\text{actual}} = \text{received} / \text{sent}$.

Gap pruning at T_{cut} : For $\rho < 100\%$, gaps older than T_{cut} are pruned by the receiver — cumulative_ack advances past abandoned symbols. Both sender (buffer eviction) and receiver (gap pruning) use the same T_{cut} , communicated at connection start and updatable mid-connection via control message.

For $\rho = 100\%$ ($T_{\text{cut}} = \infty$): no pruning. cumulative_ack advances only when gaps are filled by corrections. SACK ranges are temporary.

ACK loss handling: Cumulative ACKs are self-healing — each new ACK supersedes all previous ones. If ACK_5 is lost but ACK_6 arrives, the sender gets all information from ACK_6. No ACK retransmission needed.

6.3 Per-Symbol Delivery Outcomes

A lost symbol has three possible outcomes, depending on whether the taper function's correction symbols recover it before the cutoff T_{cut} :

Outcome 1: FEC-recovered (proactive repair arrives before T_{retx})
 Latency: L_{prop} + small delay (same as source)
 Probability: $e \times P_{\text{fec}}$

Outcome 2: ARQ-recovered (retransmit arrives after T_{retx} , before T_{cut})
 Latency: $L_{\text{prop}} + L_{\text{arq}}$ where $L_{\text{arq}} = T_{\text{retx}} + \text{RTT}/2$
 Probability: $e \times (1 - P_{\text{fec}}) \times P_{\text{arq}}$

Outcome 3: Lost (not recovered by T_{cut})
 Latency: infinity (never delivered)
 Probability: $e \times (1 - P_{\text{fec}}) \times (1 - P_{\text{arq}})$

P_{arq} is the probability that ARQ retransmission succeeds, given that FEC failed to recover the symbol. It is a conditional probability in this chain:

```

Symbol sent
|
+-- (1-e): arrives directly      -> delivered (on-time)
|
+-- e: lost on the channel
    |
    +--  $P_{\text{fec}}$ : FEC recovers it    -> delivered (on-time)
    |
    +-- (1- $P_{\text{fec}}$ ): FEC failed
        |
        +--  $P_{\text{arq}}$ : ARQ succeeds   -> delivered (late,  $L_{\text{arq}}$ )
        |
        +-- (1- $P_{\text{arq}}$ ): ARQ fails -> permanently lost
  
```

P_{arq} is derived from the reliability target ρ , not computed independently:

$$P(\text{lost}) = e \times (1 - P_{\text{fec}}) \times (1 - P_{\text{arq}}) = 1 - \rho$$

$$\text{Solving: } P_{\text{arq}} = 1 - (1 - \rho) / (e \times (1 - P_{\text{fec}}))$$

For $\rho = 100\%$: $P_{\text{arq}} = 1$. We keep retransmitting until ACK ($T_{\text{cut}} = \text{infinity}$), so ARQ eventually succeeds. FEC is still active — it handles most recoveries fast. ARQ is the backstop for what FEC misses.

For $\rho < 100\%$: $P_{\text{arq}} < 1$. We stop retransmitting at T_{cut} . Symbols not recovered by then are permanently lost.

The full delivery distribution:

$$\begin{aligned}
 P(\text{on-time delivery}) &= (1 - e) + e \times P_{\text{fec}} && \text{not lost, or FEC} \\
 P(\text{late delivery}) &= e \times (1 - P_{\text{fec}}) \times P_{\text{arq}} && \text{ARQ retransmit} \\
 P(\text{permanent loss}) &= e \times (1 - P_{\text{fec}}) \times (1 - P_{\text{arq}}) && \text{not recovered}
 \end{aligned}$$

$$\begin{aligned}
 \text{Reliability: } \rho &= 1 - P(\text{permanent loss}) \\
 \text{Tail latency: } \delta &= P(\text{late delivery}) / \rho && \text{among delivered symbols}
 \end{aligned}$$

6.4 The Triangle in Action

Under **100% reliability** ($\rho = 1$, $T_{\text{cut}} = \text{infinity}$): outcome 3 never occurs. “Tail loss from FEC” equals “tail latency events” — they are the same thing. This is the special case from Section 8.

Under **variable reliability** ($\rho < 1$, $T_{\text{cut}} < \text{infinity}$): the taper is cut off. Symbols beyond T_{cut} are permanently lost. This saves bandwidth (fewer correction symbols) and bounds latency (no recovery beyond T_{cut}), at the cost of reliability.

```

p = 100%: ----- (taper runs until ACK)
            all symbols eventually delivered

p = 98%:  -----+
            98% delivered   | T_cut
            2% lost         +-- (taper stops, accept loss)

p = 95%:  -----+
            95%             | T_cut (shorter)
            5% lost         +-- accept loss (sensor/VoIP)

```

6.5 Four-Mechanism Composition

The complete system is four independent mechanisms that compose without interfering:

Mechanism	Controls	Derives from
Copa	Total wire rate	RTT, bandwidth (delay-based)
Taper	Source/correction ratio	e , q , δ (GE + triangle)
T_{cut} + buffer	Reliability guarantee	ρ (triangle), GE B_{max}
P_{lost}	Repair/retransmit mix	ACK timing, e

Each mechanism has one job and one set of inputs. None needs to know about the others:

- Copa doesn’t know about FEC — it just limits total rate
- The taper doesn’t know about Copa — it just sets the ratio
- T_{cut} doesn’t know about the taper — it just evicts old symbols
- P_{lost} doesn’t know about T_{cut} — it just picks repair vs retransmit

The protocol hint flows through the triangle to set δ and ρ , which determine the taper amplitude and T_{cut} . Copa independently discovers the pipe capacity. P_{lost} independently tracks per-symbol loss confidence. The four mechanisms compose to produce the correct behavior for each protocol class without branching or mode switches.

7. Estimation — From Observations to Channel Parameters

7.1 What We Observe

At the sender, we receive periodic feedback:

Observation	Source	Frequency
(sent, received) per batch	ACK messages	Every batch (~10-100ms)
RTT	Echoed timestamps in ACK	Every batch
SACK ranges (out-of-order blocks)	ACK+SACK messages	Every ACK
Throughput	Delivery rate tracking	Continuous

The sender infers losses from gaps: symbols not covered by the cumulative ACK or any SACK range are presumed lost after T_{retr} has elapsed.

7.2 EWMA — Fast Point Estimate

Exponentially Weighted Moving Average of the loss rate:

$$e_hat_ewma(n) = \alpha \times (\text{lost/sent}) + (1-\alpha) \times e_hat_ewma(n-1)$$

$$\alpha = 0.1 \rightarrow \text{approximately 10-sample half-life}$$

Strengths: Simple, fast, responsive to changes.

Limitations: Single number — cannot express “confident at 2%” vs “uncertain, somewhere between 0% and 10%”. This inability to express uncertainty is why the old system needed three stacked mechanisms (EWMA + Beta margin + PI controller) to avoid under-provisioning.

7.3 Beta-Binomial Posterior — Uncertainty Quantification

The Beta distribution is the conjugate prior for Binomial observations:

```
Prior:      Beta(a, b)          (a = received, b = lost)
Update:     a' = a x decay + received
            b' = b x decay + lost
Posterior:  Beta(a', b')

Mean loss rate:  b' / (a' + b')
Variance:       a'b' / ((a'+b')2 (a'+b'+1))
Upper quantile: beta_quantile(b', a', confidence)
```

The decay factor (0.995) causes old observations to fade, allowing adaptation. This gives a half-life of approximately 138 observations ($\ln(0.5)/\ln(0.995) \approx 138$). The value is not sensitive — decay factors between 0.99 and 0.999 produce similar steady-state behavior. The choice trades off adaptation speed (lower decay = faster forgetting) against estimation stability (higher decay = smoother estimates).

Strengths: Principled uncertainty — the spread of the posterior tells us how confident we are. Tight posterior → low uncertainty → small safety margin needed.

Limitations: Cannot detect regime changes. If loss jumps from 1% to 10%, the posterior slowly drifts — it doesn’t know the old data is from a different regime.

7.4 BOCD – Regime-Aware Prediction

Bayesian Online Changepoint Detection [Adams2007] maintains a distribution over “how long since the last regime change?” (the run length).

```

Observation stream: ...000000000xxxxx000000000000000xxxx000...
                    <- regime 1 ->   <- regime 2 -> <r3>
                        ^               ^
                        changepoint   changepoint

```

Run-length distribution $P(r_t \mid \text{data})$:

Regime 1 (steady):	Changepoint:	Regime 2 (steady):
Mass at r=50	Mass splits:	Mass at r=20
(confident)	r=0 (new regime)	(confident again)
	r=51 (old continues)	
		
----- r	----- r	----- r
50	0 51	20

For each run length, BOCD maintains Beta-Binomial sufficient statistics. The predictive quantile integrates over all possible run lengths:

$$\hat{P}_{upper}(confidence) = \sum_r P(r_t = r \mid \text{data}) \times \text{beta_quantile}(\text{stats}_r, \text{confidence})$$

Key properties: - Steady state: mass concentrates at one run length $\rightarrow$ tight posterior $\rightarrow$ small margin - Changepoint: mass spreads $\rightarrow$ wide posterior $\rightarrow$ large margin (conservative) - The predictive quantile IS the safety margin — no separate PI or margin needed

7.5 GE Parameter Estimation

The GE estimator tracks transition counts with exponential decay:

```

On each symbol observation (Good or Bad):
  decay all counters by factor 0.999
  increment the appropriate transition counter:
    Good→Good:  g_to_g      Good→Bad:  g_to_b
    Bad→Good:   b_to_g      Bad→Bad:   b_to_b

```

Estimated parameters:

$$\begin{aligned}\hat{p} &= g_to_b / (g_to_g + g_to_b) & P(\text{Good} \rightarrow \text{Bad}) \\ \hat{q} &= b_to_g / (b_to_g + b_to_b) & P(\text{Bad} \rightarrow \text{Good}) \\ \hat{B} &= 1/\hat{q} & \text{mean burst length}\end{aligned}$$

Initialization: All transition counters start at zero, initial state assumed Good. The GE estimates are gated by a minimum sample count (the `is_valid()` check) — until enough transitions are observed (~50 symbols), the estimator returns default values ($B=2.0$, $q=0.5$). During this bootstrap period, the Beta prior (Section 7.3) and BOCD (Section 7.4) provide loss estimation without relying on GE burst parameters.

These estimates feed directly into the taper function shape: $\tau(t) = A \times (1 - \hat{q})^t$.

7.6 ACK Loss and Self-Healing

The feedback channel (receiver $\rightarrow$ sender) may also be lossy. If ACKs get lost, the sender lacks up-to-date knowledge of what the receiver has.

Self-healing property: Unlike a NACK-based system where a lost NACK means the sender never learns about the loss, an ACK-based system is inherently self-healing:

- If an ACK is lost, the sender simply does not advance its knowledge of receiver state. Symbols remain in the retransmit buffer.
- After T_{retx} elapses without acknowledgment, the sender retransmits — this is correct behavior whether the original was lost or just the ACK was lost.
- If the original arrived but the ACK was lost, the receiver gets a duplicate (harmless — deduplicated by sequence number) and sends another ACK.
- Eventually an ACK gets through, and the sender's state catches up.

ACK lost scenario:

```

Sender:   [S1] [S2] ... wait  $T_{\text{retx}}$  ... [S1'] ... receives ACK ... done
                                     ↑
                                     retransmit (safe: either original
                                     or ACK was lost, both handled)

```

No separate mechanism needed — the retransmit timeout handles both cases.

This eliminates the need for RX path loss estimation, NackAck echo, and NACK effectiveness tracking. The system is simpler and more robust.

7.7 Estimation Error and Overhead

Estimation error directly maps to overhead:

If $e_{\text{hat}} > e_{\text{true}}$: over-provisioning → wasted bandwidth
 If $e_{\text{hat}} < e_{\text{true}}$: under-provisioning → more ARQ latency events

Overhead gap = $(e_{\text{hat}} - e_{\text{true}}) / e_{\text{true}}$

BOCD minimizes this gap by adapting the estimation confidence to the regime: - Steady state: $\hat{\epsilon} \approx \epsilon_{\text{true}}$ (tight posterior) - Transition: $\hat{\epsilon} > \epsilon_{\text{true}}$ (conservative, correct behavior)

8. The Optimization Problem

8.1 Formal Statement

minimize: $r = A/q$ (correction rate = bandwidth cost)

subject to: $e \times (1 - P_{\text{fec}}(A, q)) \leq \delta$ (tail latency constraint)

where: $\tau(t) = A \times (1-q)^t$ (taper function)
 P_{fec} depends on A, q, e, W (FEC recovery probability)

Input: δ (tail latency target, from protocol hint)

Output: A^* (optimal taper amplitude), $r^* = A^*/q$ (optimal correction rate)

8.2 Corrected Model (canonical)

Let's reconsider. The taper generates correction symbols at known positions. For a window of size W , the taper generates exactly $r \times W$ correction symbols total. The question is: given that a source symbol is lost, how many correction symbols covering it will arrive at the receiver?

In a sliding window code, repair symbols are linear combinations of all source symbols in the window. A repair at offset t from source s covers s as long as s is still in the window ($t < W$).

Number of correction symbols generated while s is in the window:

$$N_{\text{correction}} = \sum_{t=0}^{W-1} \tau(t)$$

Of these, each survives independently with probability $(1-\epsilon)$. Expected arrivals:

$$R = N_{\text{correction}} \times (1-\epsilon) = (A/q) \times (1 - (1-q)^W) \times (1-\epsilon)$$

For the decoder to recover s , we need at least 1 linearly independent repair (with systematic code, if all other source symbols arrived, one repair suffices).

But the actual requirement is more nuanced: we need as many repair symbols as there are lost source symbols in the window. If k symbols are lost in the window, we need k repair arrivals.

Average case: In a window of size W with loss rate ϵ , expected losses $= \epsilon W$. Expected repair arrivals $= r \times W \times (1-\epsilon)$. For recovery: $r \times W \times (1-\epsilon) \geq \epsilon W$, giving $r \geq \epsilon/(1-\epsilon) = r_{\text{IT}}$. This confirms the IT minimum.

Tail case: We need recovery even when more symbols are lost than average. The number of losses in a window follows a distribution determined by the GE model. The tail latency constraint requires recovery at the δ -th percentile.

Let K be the number of lost symbols in a window. For the GE model:

$$\begin{aligned} P(K = k) &\text{ depends on the burst structure} \\ \text{For large } W: K &\approx e \cdot W \text{ with variance determined by burstiness} \end{aligned}$$

For recovery: number of arriving repairs $\geq K$.

Repairs arrive $= \text{Binomial}(r \times W, 1-\epsilon)$, approximately $\text{Normal}(rW(1-\epsilon), rW(1-\epsilon)\epsilon)$.

Constraint: $P(\text{repairs} < K) \leq \delta$.

Using normal approximation:

$$P(\text{repairs} < K) \approx P(\text{Normal}(rW(1-\epsilon), rW(1-\epsilon)\epsilon) < K)$$

This requires the δ -quantile of repairs to exceed the $(1-\delta)$ -quantile of losses. The algebra gives:

$$r \geq e/(1-\epsilon) + z_{\delta} \times \sqrt{(e/(W(1-\epsilon)))}$$

where z_{δ} is the standard normal quantile [Abramowitz1964] for probability δ .

The explicit P_{fec} formula. Given correction rate r , loss rate e , burst variance $s2_{\text{burst}}$, and window size W :

$$\begin{aligned} \text{Losses in window: } K &\sim \text{Normal}(We, We(1-\epsilon)s2_{\text{burst}}) \\ \text{Surviving corrections: } C &\sim \text{Normal}(rW(1-\epsilon), rWe(1-\epsilon)) \end{aligned}$$

$$P_{\text{fec}} = P(C \geq K) = \Phi(z)$$

$$\text{where } z = \sqrt{W} \times (r(1-\epsilon) - e) / \sqrt{e(1-\epsilon)(r + s2_{\text{burst}})}$$

At the IT minimum ($r = e/(1-\epsilon)$): $z = 0$, $P_{\text{fec}} = 0.5$ (coin flip — half the time corrections suffice, half the time they don't). The margin term in Section 8.4's r^* pushes z positive, giving $P_{\text{fec}} > 0.5$.

Note: The r^* formula from Section 8.4 is a first-order approximation of inverting this P_{fec} formula. It drops r from the variance denominator (assumes $r \ll s2_{\text{burst}}$). For practical r values (0.05-0.25) this is accurate. For exact computation, invert the P_{fec} formula numerically.

This normal approximation is valid when $W \gg B$ (window much larger than mean burst) — which covers all practical scenarios ($W=50$, $B=2-3$). For edge cases where $W \approx B$, the exact GE distribution is computable via transfer matrix dynamic programming in $O(W^2)$. See Appendix D item 4.

8.3 Burst Variance Correction

The normal approximation in 8.2 assumes iid losses (Binomial variance). On a GE channel, losses are correlated — bursts inflate the variance.

The GE autocorrelation decays with eigenvalue $(1-p-q)$. The variance of losses in a window of size W is:

$$\begin{aligned} \text{Var}_{\text{iid}}(K) &= W \times e \times (1-e) && \text{(independent losses)} \\ \text{Var}_{\text{GE}}(K) &= W \times e \times (1-e) \times \sigma^2_{\text{burst}} && \text{(burst-correlated losses)} \\ \sigma^2_{\text{burst}} &= 1 + 2(1-p-q)/(p+q) && \text{(variance inflation factor)} \end{aligned}$$

Scenario	$p+q$	σ^2_{burst}	Meaning
DC	0.501	3.0	3× wider variance than iid assumption
WiFi	0.513	2.9	similar
LTE	0.42	3.8	significant inflation
Satellite	0.33	5.1	iid approximation seriously wrong

We compute σ^2_{burst} directly from the GE estimator's $\hat{p}$ and $\hat{q}$.

8.4 The Corrected Optimal Correction Rate

$$r^* = \frac{e}{(1-e)} + \frac{z_{\delta} \times \sqrt{e \times s2_{\text{burst}} / (W \times (1-e))}}{\sqrt{1-e}}$$

IT minimum tail margin
(accounts for burst correlation)

$$s2_{\text{burst}} = 1 + 2(1-p-q)/(p+q)$$

$$z_{\delta} = \text{standard normal quantile for } (1-\delta)$$

Properties: - IT minimum $\varepsilon/(1-\varepsilon)$ is the dominant term - Tail margin scales as $1/\sqrt{W}$ — larger windows need proportionally less margin - z_{δ} controls the margin: tighter $\delta \rightarrow$ larger $z_{\delta} \rightarrow$ more margin - σ^2_{burst} amplifies the margin for bursty channels (large for small $p+q$)

Note: This formula uses the raw loss rate e . For the canonical production formula including codec overhead, replace e with $e_{\text{hat}} = e + e_{\text{codec}} \times (1-(1-e)^W)$ from Section 9.2. The codec-adjusted version accounts for decoder invocation probability on systematic codes.

8.5 Worked Examples

Using z_{δ} values: $z(1e-2) = 2.33$, $z(1e-4) = 3.72$, $z(1e-6) = 4.75$

The margin term is: $z_{\delta} \times \sqrt{(\epsilon \times \sigma^2_{\text{burst}} / (W \times (1-\epsilon)))}$

DC ($\epsilon=0.001$, $W=50$, $\sigma^2_{\text{burst}}=3.0$):

Bulk ($\delta=1e-2$): $r^* = 0.001 + 2.33 \times \sqrt{(0.001 \times 3.0 / 49.95)} = 0.1\% + 1.8\% = 1.9\%$
Auto ($\delta=1e-4$): $r^* = 0.001 + 3.72 \times \sqrt{(0.001 \times 3.0 / 49.95)} = 0.1\% + 2.9\% = 3.0\%$
Realtime ($\delta=1e-6$): $r^* = 0.001 + 4.75 \times \sqrt{(0.001 \times 3.0 / 49.95)} = 0.1\% + 3.7\% = 3.8\%$

WiFi ($\epsilon=0.025$, $W=50$, $\sigma^2_{\text{burst}}=2.9$):

Bulk ($\delta=1e-2$): $r^* = 2.6\% + 2.33 \times \sqrt{(0.025 \times 2.9 / 48.75)} = 2.6\% + 2.8\% = 5.4\%$
Auto ($\delta=1e-4$): $r^* = 2.6\% + 3.72 \times \sqrt{(0.025 \times 2.9 / 48.75)} = 2.6\% + 4.5\% = 7.1\%$
Realtime ($\delta=1e-6$): $r^* = 2.6\% + 4.75 \times \sqrt{(0.025 \times 2.9 / 48.75)} = 2.6\% + 5.8\% = 8.4\%$

Satellite ($\epsilon=0.09$, $W=50$, $\sigma^2_{\text{burst}}=5.1$):

Bulk ($\delta=1e-2$): $r^* = 9.9\% + 2.33 \times \sqrt{(0.09 \times 5.1 / 45.5)} = 9.9\% + 7.4\% = 17.3\%$
Auto ($\delta=1e-4$): $r^* = 9.9\% + 3.72 \times \sqrt{(0.09 \times 5.1 / 45.5)} = 9.9\% + 11.8\% = 21.7\%$
Realtime ($\delta=1e-6$): $r^* = 9.9\% + 4.75 \times \sqrt{(0.09 \times 5.1 / 45.5)} = 9.9\% + 15.1\% = 25.0\%$

8.6 Three-Variable Optimization

Section 8.4 solves for r^* given δ (with $\rho=100\%$). Here we generalize to all three modes of the bandwidth/latency/reliability triangle.

Taper with cutoff

When $\rho < 100\%$, the taper is truncated at T_{cut} :

$$\begin{aligned} \tau(t) &= A \times (1-q)^t & \text{for } t \leq T_{\text{cut}} \\ \tau(t) &= 0 & \text{for } t > T_{\text{cut}} \end{aligned}$$

Total correction rate with cutoff:

$$r = A \times \sum_{t=0}^{T_{\text{cut}}} (1-q)^t = A \times (1 - (1-q)^{T_{\text{cut}}+1}) / q$$

For $T_{\text{cut}} = \infty$ ($\rho = 100\%$): reduces to $r = A/q$ (Section 4.3).

Mode 1: Given (δ, ρ) $\rightarrow$ compute r

Fix tail latency target δ and reliability ρ . Compute minimum bandwidth r^* .

Step 1: From ρ , find T_{cut} . The reliability $\rho = P(\text{recovered within } T_{\text{cut}})$. Using the corrected model (Section 8.4):

$$T_{\text{cut}} \text{ such that: } e \times (1 - P_{\text{fec}}(T_{\text{cut}})) \times (1 - P_{\text{arq}}(T_{\text{cut}})) = 1 - \rho$$

For $\rho = 100\%$: $T_{\text{cut}} = \infty$ (no finite solution — see Section 6.4). For $\rho < 100\%$: solve via binary search. $P(\text{recovered by } T_{\text{cut}})$ is monotone increasing in T_{cut} (more time = more corrections = higher recovery):

Algorithm: find T_{cut} from ρ

```

lo = 0
hi = W x 10                                (upper bound: many window lengths)
while hi - lo > tolerance:
    mid = (lo + hi) / 2
    corrections = A x (1 - (1-q)^(mid+1)) / q x (1-e)
    P_recovered = Phi(sqrt(W) x (r(1-e)-e) / sqrt(e(1-e)(r+s2_burst)))    (Section
8.2)
    if P_recovered <  $\rho$ :
        lo = mid                            (need more time)
    else:
        hi = mid                            (enough time)
T_cut = hi

```

Convergence is guaranteed because $P(\text{recovered})$ is monotone in T_{cut} . Typically converges in ~ 20 iterations ($\log_2$ of search range).

Step 2: From δ , find A using the tail latency constraint (Section 8.4):

A^* such that: $e \times (1 - P_{\text{fec}}(A^*)) \leq \delta$ (among delivered symbols)

Step 3: Compute $r^* = A^* \times (1 - (1-q)^{\{T_{\text{cut}}+1\}}) / q$.

Special case $\rho = 100\%$: $T_{\text{cut}} = \infty$, $r^* = A^*/q = \varepsilon/(1-\varepsilon) + z_{\delta}\sqrt{\dots}$. This is the formula from Section 8.4.

Mode 2: Given $(r, \rho) \rightarrow \text{compute } \delta$

Fix bandwidth r and reliability ρ . Compute resulting tail latency δ .

```

From  $\rho$ : find  $T_{\text{cut}}$  (same as Mode 1, Step 1)
From  $r$  and  $T_{\text{cut}}$ :  $A = r \times q / (1 - (1-q)^{\{T_{\text{cut}}+1\}})$ 
From  $r$  and  $W$ :  $P_{\text{fec}} = \text{Phi}(\text{sqrt}(W) \times (r(1-e)-e) / \text{sqrt}(e(1-e)(r+s2\_burst)))$ 
(Section 8.2)
Result:  $\delta = e \times (1 - P_{\text{fec}}) \times P_{\text{arq}} / \rho$ 

```

Mode 3: Given $(r, \delta) \rightarrow \text{compute } \rho$

Fix bandwidth r and tail latency δ . Compute resulting reliability ρ .

```

From  $r$ :  $A = r \times q / (1 - (1-q)^{\{T_{\text{cut}}+1\}})$  (depends on  $T_{\text{cut}}$ )
From  $\delta$ : determine how much of the taper is "on-time" vs "late"
From  $A$  and the taper integral:  $\rho = \text{total recovery probability within } T_{\text{cut}}$ 

```

This mode requires solving for T_{cut} and A simultaneously (they're coupled). Binary search on T_{cut} works: ρ is monotone in T_{cut} (larger T_{cut} = more recovery time = higher ρ), and r is monotone in T_{cut} (larger T_{cut} = more corrections needed = higher r). For a given r budget:

Algorithm: find ρ given (r, δ)

```

lo = 0
hi = W x 10
while hi - lo > tolerance:
    mid = (lo + hi) / 2
    A = r x q / (1 - (1-q)^(mid+1))
    P_fec = Phi(sqrt(W) x (r(1-e)-e) / sqrt(e(1-e)(r+s2_burst))) (Section 8.2)
    if A produces rate > r:
        hi = mid (T_cut too large for budget)
    else:
        lo = mid (can afford more T_cut)
T_cut = lo
ρ = P(recovered within T_cut) (from taper integral)

```

Convergence guaranteed by monotonicity. The resulting ρ is the maximum reliability achievable within the bandwidth budget r at tail latency δ .

Worked examples

Example 1: Bulk file transfer (WiFi, $\epsilon=0.025$)

Fix: $\rho = 100\%$, minimize r
 Compute: δ (tail latency)

$r^* = 2.6\% + z_{\delta} \times 1.2\%$ (from Section 8.5, WiFi row)

At minimum $r = r_{IT} = 2.6\%$: $\delta = e = 2.5\%$ (every lost symbol goes to ARQ)
 Tail latency $\approx T_{retx} + RTT/2$ for 2.5% of symbols
 For $RTT = 50\text{ms}$: $L_{arq} \approx 100\text{ms}$ for 2.5% of symbols

To get $\delta = 1e-2$: $r^* = 5.4\%$ (from worked examples)

Example 2: VoIP (WiFi, $\epsilon=0.025$)

Fix: $\delta = 150\text{ms}$ budget $\rightarrow T_{cut} = 150\text{ms} / \text{symbol_time}$
 $r = 5\%$ (codec + small overhead)
 Compute: ρ (reliability)

With $r = 5\%$, the taper generates enough correction symbols that:

- $P_{fec} \approx 0.90$ (90% of lost symbols FEC-recovered, zero latency)
- P_{arq} within 150ms ≈ 0.08 (8% recovered by retransmit)
- $P_{lost} = 0.02$ (2% permanently lost)
- $\rho = 98\%$

The VoIP codec conceals the 2% frame loss.

Example 3: Live video (WiFi, $\epsilon=0.025$)

Fix: $\delta = 33\text{ms}$ (one frame at 30fps), $\rho = 99.9\%$
 Compute: r (bandwidth)

Need 99.9% of symbols delivered within 33ms.
 T_{cut} determined by $\rho = 99.9\%$: $T_{cut} \approx 3 \times RTT$
 A determined by δ : need $P(\text{recovery within } 33\text{ms}) \geq 0.999$
 $r^* \approx 8.4\%$ (close to Realtime from Section 8.5)

Example 4: Gaming (LTE, $\epsilon=0.05$)

Fix: $\delta = 20\text{ms}$ (tight), $\rho = 99\%$ (1% loss acceptable)
 Compute: r (bandwidth)

Very tight latency + moderate reliability $\rightarrow$ aggressive FEC
 $T_{\text{cut}} \approx 2 \times \text{RTT}$ (short: accept 1% loss)
 $r^* \approx 12\%$ (most budget goes to proactive FEC within 20ms)

Example 5: Sensor telemetry (Satellite, $\epsilon=0.09$)

Fix: $r = 5\%$ (minimal bandwidth), $\rho = 95\%$
 Compute: δ (tail latency)

Low bandwidth budget + high loss $\rightarrow$ many symbols go to ARQ
 T_{cut} determined by $\rho = 95\%$
 $\delta \approx 15\%$ (15% of delivered symbols arrive late)
 Acceptable for periodic sensor readings.

9. Codec Overhead Integration

9.1 Decoder Invocation Probability

For systematic codes (METTLE, RLC, RaptorQ), the decoder is only invoked when at least one source symbol in the window is lost:

$$P(\text{decoder invoked}) = 1 - (1-e)^W$$

Scenario	ϵ	W=50	W=200
DC	0.001	4.9%	18.1%
WiFi	0.025	72.1%	99.4%
Satellite	0.09	99.1%	$\approx 100\%$

9.2 Effective Codec Overhead

The codec overhead (ϵ_{codec}) represents extra symbols the decoder needs beyond the information-theoretic minimum:

Backend	ϵ_{codec}	Reason
Reed-Solomon	0.0%	MDS: any k of n suffices
RLC	0.4%	Near-MDS: rare rank deficiency [RFC8681]
RaptorQ	1.0%	Fountain code convergence overhead [RFC6330]
METTLE	15.0%	Sparse random matrix: needs more eqns [Yu2025]
Streaming	0.0%	Rate-optimal by construction [Badr2017]

Weighted by decoder invocation probability:

$$e_{\text{codec_eff}} = e_{\text{codec}} \times P(\text{decoder invoked}) = e_{\text{codec}} \times (1 - (1-e)^W)$$

The corrected correction rate becomes:

$$r^* = (e + e_{\text{codec_eff}}) / (1-e) + z_{\delta} \times \sqrt{((e + e_{\text{codec_eff}}) / (W \times (1-e)))}$$

9.3 Impact on METTLE at DC

Without weighting: r^* includes 15% codec overhead $\rightarrow$ 16.1% correction rate. With weighting: $\epsilon_{\text{codec_eff}} = 0.15 \times 0.049 = 0.74\% \rightarrow$ 1.8% correction rate.

The weighting reduces METTLE's DC overhead by $\sim 9\times$.

10. Multi-Protocol Extension

10.1 Per-Symbol Latency Classes

Different traffic types have different δ targets:

Symbol s has latency class $c(s)$ with target $\delta_{\{c(s)\}}$

10.2 Interleave Before vs After FEC

After FEC (separate streams):

Realtime packets $\rightarrow$ [FEC encoder ($\delta=1e-6$)] $\rightarrow$ channel
Bulk packets $\rightarrow$ [FEC encoder ($\delta=1e-2$)] $\rightarrow$ channel

Each stream has its own taper. No correction sharing. Simple.

Before FEC (shared stream):

All packets $\rightarrow$ [mixed FEC encoder] $\rightarrow$ channel

One taper covers everything. Repair symbols are linear combinations of ALL source symbols [RFC8681] — a repair can recover ANY lost symbol regardless of class.

Advantage of shared: repair symbols are fungible. A repair generated “for” a Realtime symbol can recover a Bulk symbol if needed. Total correction budget can be lower than the sum of separate budgets (statistical multiplexing).

Disadvantage of shared: the taper must be sized for the tightest class. If 1% of traffic is Realtime ($\delta=1e-6$) and 99% is Bulk ($\delta=1e-2$), you pay Realtime-level overhead on everything.

10.3 When Shared Wins

Shared FEC is cheaper when the traffic mix is balanced or dominated by the tight class. Separate streams are cheaper when the tight class is a small fraction. The crossover depends on the specific δ values and loss rate.

Decision rule: Compare total correction bandwidth:

$\text{shared_cost} = r^*(e, \min(\delta_c))$ (one encoder, tightest δ)
 $\text{separate_cost} = \sum_c f_c \times r^*(e, \delta_c)$ (per-class, weighted by fraction f_c)

Choose whichever is lower.

10.4 Extending the Formula

For shared FEC with per-symbol δ , the constraint becomes:

For each class c : $e \times (1 - P_{\text{fec}}) \leq \delta_c$

Since P_{fec} is the same for all symbols (shared repair), the binding constraint is the tightest: $\delta_{\min} = \min(\delta_c)$. The formula reduces to the single-class case with $\delta = \delta_{\min}$.

11. Verification

11.1 Simulation Approach

Generate synthetic loss traces from the GE model, apply the taper function, and measure whether the actual tail latency matches the theoretical prediction.

For each trial:

1. Generate GE loss trace: {lost_1, lost_2, ..., lost_N}
2. Generate correction schedule from taper: {correction_1, correction_2, ...}
3. Apply channel loss to both source and correction symbols
4. For each lost source symbol:
 - a. Check if enough repair symbols arrived (FEC recovery)
 - b. If not, mark as ARQ-recovered (retransmit needed)
5. Count ARQ events (symbols not FEC-recovered)
6. Measure: $\text{actual_arq_fraction} = \text{arq_events} / N$

Over many trials:

Verify: $P(\text{actual_arq_fraction} > \delta)$ is small

Verify: mean correction rate $\approx r^*$

11.2 Analytical Predictions to Verify

Prediction	Formula	Test
IT minimum dominates at high loss	$r^* \approx \epsilon/(1-\epsilon)$ when W large	Satellite scenario
Tail margin scales as $1/\sqrt{W}$	$r(W=200) < r(W=50)$	Compare window sizes
Taper shape matches GE	Decay rate = $\hat{q}$ from estimator	Compare simulated vs theoretical
Codec overhead weighting	METTLE DC overhead $\sim 0.74\%$ not 15%	DC scenario with METTLE
Protocol hint only affects δ	Realtime(1e-5) = Auto(1e-7)	Same estimator, different hints

11.3 Boundary Cases

Case	Expected behavior	Why
$\epsilon = 0$ (no loss)	$r^* = 0$	No correction needed
$\epsilon \rightarrow 1$ (total loss)	$r^* \rightarrow \infty$	Can't recover anything with FEC alone
$\delta = \epsilon$ (every loss to ARQ)	$r^* = 0$	No FEC needed, all ARQ
$\delta \rightarrow 0$ (zero ARQ tolerance)	$r^* \rightarrow \infty$	Must FEC-recover everything
$W = 1$ (no window)	Margin term large	Single-symbol recovery needs more redundancy
$q = 1$ (no burst memory)	$\tau(t) = \text{flat}$	Reduces to iid case

11.4 Connection to Existing Benchmarks

The bench_suite already measures overhead_pct and recovery_pct per scenario. To verify the model:

1. Compute r^* from the formula for each scenario's (ϵ , δ , W)
2. Compare with the bench_suite's measured overhead
3. The gap between r^* and measured overhead = estimation tax + implementation overhead

4. Track this gap across benchmark runs — it should decrease as we improve the implementation

12. Congestion Control Integration

12.1 Why Delay-Based CC is Required

On lossy links (WiFi, LTE, satellite), loss-based CC (NewReno, CUBIC) is catastrophic. Every random channel loss triggers cwnd halving, collapsing throughput on exactly the links raptorpath is designed for:

NewReno on 2.5% WiFi loss:

- > cwnd halves on every loss event
- > throughput oscillates wildly, average $\ll$ capacity
- > our FEC tries to compensate -> more traffic -> more congestion

Delay-based CC on 2.5% WiFi loss:

- > loss + stable RTT = channel loss -> ignore, let FEC handle it
- > loss + rising RTT = congestion -> reduce rate
- > throughput stays near capacity

Delay-based CC distinguishes congestion (queue buildup = rising RTT) from channel loss (random drops = stable RTT). This is essential for raptorpath.

12.2 QUIC Datagrams Bypass Quinn's CC

Raptorpath sends symbol data as QUIC unreliable datagrams, which bypass Quinn's built-in congestion control (NewReno). Our own CC is the sole rate limiter for data traffic. Quinn's CC only applies to QUIC streams (handshake, reliable control messages — small and infrequent).

This means our CC is solely responsible for not flooding the network.

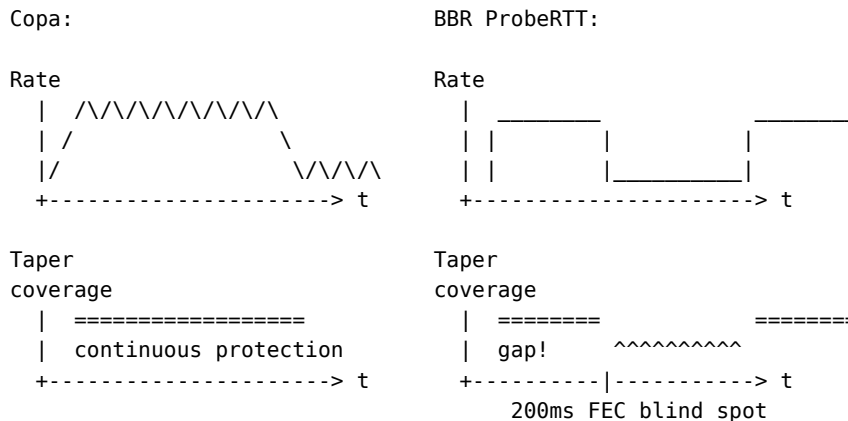
12.3 Copa vs BBR

The current implementation uses BBR (ADR-0019). We recommend migrating to Copa [Copa2018] because of better interaction with the taper function:

Aspect	BBR	Copa
Core idea	measure $\max_bw \times \min_rtt = BDP$	$rate = 1/(d \times dq)$
Queue draining	ProbeRTT: 200ms forced drain every 10s	Natural oscillation
Phases	Startup -> ProbeBw -> ProbeRtt (state machine)	No phases, one formula
Lossy links	RTT-trend check to ignore random loss	Delay-based, same effect
Complexity	~300 lines, multiple edge cases	One rate formula
Taper interaction	ProbeRTT creates FEC protection gap	Smooth, no gaps

Implementation status: Copa is recommended as the target congestion control algorithm. The current codebase uses BBR (ADR-0019). Migration to Copa is a future implementation task. The model and formulas in this paper are CC-agnostic — they work with any delay-based CC that provides a total sending rate.

The critical difference — taper compatibility:



Copa's rate oscillation is smooth and fast (order of RTT). The taper function uses $RATIO$ (corrections per source symbol), which stays constant. Absolute rates oscillate gently, but the EWMA smooths throughput estimates. The taper doesn't notice Copa's oscillation.

BBR's ProbeRTT drops $cwnd$ to 4 symbols for 200ms. During this period:

- Source rate drops to nearly zero
- Correction rate drops proportionally (ratio constant, absolute count tanks)
- Source symbols sent during ProbeRTT get almost no FEC protection
- If a burst loss hits during ProbeRTT, recovery is severely degraded
- After ProbeRTT exits: data burst -> potential queue buildup

12.4 Copa's Rate Formula

Copa targets a queue occupancy that balances throughput and delay:

$$rate = 1 / (d_copa \times dq)$$

where:

d_copa = Copa parameter (controls target queue depth; default 0.5)

dq = queuing_delay = $RTT_current - RTT_min$

When dq is small (queue empty): rate is high -> fill the pipe. When dq is large (queue building): rate drops -> drain the queue.

This naturally oscillates: send fast -> queue builds -> send slow -> queue drains -> send fast again. The oscillation frequency is $\sim 1/RTT$ and amplitude depends on d_copa . No periodic forced drain phase needed.

Copa's d_copa is NOT the same as our tail latency target δ . Copa's d_copa controls the target queue depth at the bottleneck ($1/d_copa$ packets). Our δ controls the tail probability of late delivery. They are independent parameters that happen to use similar notation in their respective papers.

$d_copa = 0.5$ (from the Copa paper [Copa2018]) targets a queue of 2 packets — very shallow. The Copa+ follow-up paper explores adaptive d_copa , which could be coupled to the protocol hint: Realtime (tight latency) -> higher d_copa (shallower queue, ~ 1 less packet of delay) and Bulk (throughput) -> lower d_copa (deeper queue, better link utilization). However, the gain is approximately 1 packet of queue delay ($\sim 1ms$ at typical speeds) — negligible compared to RTT. The default $d_copa = 0.5$ is sufficient for all scenarios.

min_rtt estimation: Copa uses the minimum observed RTT in a 10-second sliding window (same duration as BBR's min_rtt window). Copa's natural rate oscillation periodically reduces the queue to near-empty, refreshing the minimum within this window. If min_rtt has not been refreshed for an extended period (>20s), the system can force a brief rate reduction as a fallback — though this is rarely needed because Copa's oscillation provides natural refreshing.

12.5 CC + Taper: The Complete Architecture

Copa determines: total_rate (symbols/sec on the wire)

Taper determines: r* (correction symbols per source symbol)

$$\text{source_rate} = \text{total_rate} / (1 + r^*)$$
$$\text{correction_rate} = \text{total_rate} \times r^* / (1 + r^*)$$

When channel worsens: e rises -> r* rises -> source_rate falls

When congestion: dq rises -> total_rate falls -> both fall

When channel clears: e drops -> r* drops -> source_rate rises

The two controllers are orthogonal: - Copa controls HOW FAST (total symbols per second) - Taper controls HOW MUCH redundancy (correction per source ratio)

Neither needs to know about the other. Copa sees total traffic; taper sees loss rate. They compose naturally.

12.6 ECN as Opportunistic Enhancement

If the network path supports ECN [RFC3168], congestion is signaled by router marking (CE bit) instead of dropping. This provides: - Congestion detection without loss -> even better for delay-based CC - Positive identification: marked = congestion, dropped = channel loss - No need to distinguish via RTT trends (direct signal)

QUIC validates ECN support at connection startup. If supported, use it. If not (common on wireless), fall back to Copa's delay-based detection.

12.7 Application Back-Pressure

When r* is so high that source_rate = total_rate/(1+r*) drops below the application's minimum (e.g., VoIP codec needs 64kbps), the system signals back-pressure: "the channel cannot support your required quality at this loss rate."

The application must respond: - Reduce quality (lower bitrate codec, lower video resolution) - Accept lower reliability (increase d, decrease p) - Wait for better conditions

This is a resource allocation problem, not a CC problem. The CC works correctly; it's the application that needs to adapt.

Back-pressure mechanism: The sender stops reading from the source (TUN interface, application socket) when the retransmit buffer reaches buffer_max. This causes the kernel's write buffer to fill, which causes the application's write() to block — identical to TCP behavior when the send buffer is full.

No special API is needed for basic operation: applications using the TUN interface experience standard blocking write() semantics. Data is accepted when the channel can handle it and blocked when it can't.

Optional stats API (for advanced applications): Applications that want finer control can subscribe to channel statistics: - Per-path: ϵ , RTT, throughput, correction rate r , Copa rate - Aggregate: effective reliability ρ , tail latency δ , correction deficit - Events: backpressure start/end, path added/removed, regime change

The API also allows overriding protocol characteristics mid-connection: - Adjust δ (tail latency target) or ρ (reliability target) - Change T_{cut} (affects eviction and gap pruning) - Force path preferences

This API is implementation-specific and not part of the core model.

13. Multi-Path Scheduling

13.1 Why FEC Beats MPTCP

MPTCP (Multi-Path TCP) schedulers are fundamentally limited by **head-of-line (HOL) blocking**: TCP requires in-order delivery, so a packet on the slow path blocks all fast-path packets at the receiver. MPTCP schedulers (round-robin, weighted, BLEST [Ferlin2016]) try to minimize this by avoiding slow paths, but they can't eliminate it.

Our FEC-based model is fundamentally different:

MPTCP:	Raptorpath:
Path A: [P1] [P2] [P5] [P6]	Path A: [S1] [C] [S3] [C]
Path B: [P3] [P4]	Path B: [S2] [S4] [C]
Receiver must wait for P3 before delivering P4, P5, P6. HOL blocking on slow path.	Decoder needs ANY k of n symbols. Order doesn't matter. No HOL blocking.

The decoder doesn't care WHICH symbols arrive — just HOW MANY. A repair symbol on Path A can recover a lost source on Path B. Source symbols can arrive in any order. The reorder buffer handles sequencing after decode.

13.2 Per-Path Model

Each path i runs independently with its own:

Copa _{i} :	$\text{rate}_i = 1 / (d \times dq_i)$	total sending rate
GE _{i} :	(e_i, p_i, q_i)	loss model
Taper _{i} :	$\tau_i(t) = A_i \times (1 - q_i)^t$	correction density
r _{i} :	correction rate = A_i / q_i	source/correction ratio

All paths share: - **One source stream**: source symbols are distributed across paths - **One retransmit buffer**: any path can retransmit any un-ACKed symbol - **One FEC encoder window**: repair symbols cover the same source window

13.3 Unified Symbol Stream and Interleaving

Each path carries a **unified stream** of source + correction symbols, interleaved at that path's own taper ratio. The interleaving is essential for burst protection — corrections scattered among source symbols survive bursts that wipe out consecutive symbols:

Path A ($e=0.05$, $r=0.05$): [S][S][S][S][C][S][S][S][C]... 5% corrections
 Path B ($e=0.10$, $r=0.11$): [S][S][C][S][S][C][S][S][C]... 11% corrections

During burst on Path A: [S][X][X][X][C][S]...
 lost ^ this correction survives!
 -> decoder can recover

Source and corrections must NOT be separated between paths. If source goes on one path and corrections on another, the source path has no interleaved corrections, and burst protection fails:

BAD: source/correction separation
 Path A: [S][S][S][S][S][S]... <- burst wipes everything, no protection
 Path B: [C][C][C][C][C][C]... <- corrections exist but on wrong path

The scheduler distributes the combined stream across paths. Each path independently interleaves at its own ratio.

13.4 Correction Deficit

The **correction deficit** is the total expected corrections still needed across all paths:

$$\text{deficit} = \text{SUM}_{\{s \text{ in un-ACKed}\}} e_s$$

where e_s = channel loss rate of path(s) at the time symbol s was sent

Each source or correction symbol sent on path i adds e_i to the deficit (it might be lost). Each ACKed symbol removes its send-time e_s (confirmed survived).

Why $e/(1-e)$ is a geometric series: When a correction symbol is itself lost, it needs replacement. This creates a chain:

Source lost: 0.30 (30% loss)
 Corrections also lost: 0.30 x 0.30 = 0.09
 Replacements also lost: 0.30^3 = 0.027
 ...
 Total = 0.30 + 0.09 + 0.027 + ... = 0.30 / (1 - 0.30) = 0.4286

This is why $r_{IT} = e/(1-e)$, not just e . The $(1-e)$ denominator accounts for the infinite chain of correction-of-correction loss. The deficit counter captures this naturally: lost corrections add to the deficit, generating more corrections, until enough survive.

Cross-path correction deficit: When source is on path A (e_A) and corrections go on path B (e_B), the formula becomes:

$$r = e_A / (1 - e_B)$$

This emerges from the deficit dynamics: source adds e_A , each surviving correction on path B removes $(1-e_B)$. Equilibrium: $e_A = r \times (1-e_B)$.

13.5 Effective Delivery Time and Bandwidth

For each path i :

E_i	=	$RTT_i/2 + e_i \times t_{\text{recovery}_i}$	effective delivery time [sec]
B_{eff_i}	=	$C_i / (1 + r_i)$	source-carrying capacity [sym/s]
e_{combined}	=	$\text{SUM}(C_i \times e_i) / \text{SUM}(C_i)$	throughput-weighted loss [prob]

where $t_{\text{recovery}_i} = P_{\text{fec}_i} \times t_{\text{fec}_i} + (1 - P_{\text{fec}_i}) \times L_{\text{arq}_i}$ is the expected recovery time on path i if the symbol is lost. t_{fec_i} is the FEC recovery time (Section 3.4) and L_{arq_i} is

the ARQ recovery latency on that path. The delay spectrum concept [Facenda2022] provides related per-link rate allocation theory for multi-path streaming.

13.6 Scheduler Ratio Adjustment

The scheduler can adjust the per-path source/correction ratio to favor certain paths for source symbols. This is only beneficial for **latency-sensitive** traffic (source symbols are immediately processable, corrections are not — so putting more source on a fast path saves latency):

```
Default (natural taper):
  Path A (fast, e=0.05): [S][S][S][S][C][S][S][S][S][C]...   r=0.05

Latency-optimized (scheduler shifts source to fast path):
  Path A (fast):        [S][S][S][S][S][S][S][S][S][C]...   r'=0.02
  Path B (slow):        [S][C][C][S][C][C][S][C][C]...       r'=0.30
                                                                (absorbs deficit)
```

For **bandwidth-optimized** traffic: no adjustment needed. Source and corrections cost the same bandwidth. The natural taper ratio is optimal.

Constraint: Each path must maintain minimum interleaving for burst protection. The scheduler cannot reduce the correction ratio below a floor that leaves the path unprotected during bursts.

13.7 Burst Protection During Ratio Adjustment

When the scheduler reduces a path's correction ratio (more source, fewer corrections), that path has fewer same-path corrections to survive a burst. But burst protection is already handled by two existing mechanisms:

Global correction deficit (Section 13.4): When one path's corrections are reduced, the deficit grows — and other paths absorb it by generating more corrections. The total correction budget across all paths remains matched to the total loss. No per-path floor is needed because the correction budget is global, not per-path.

Cross-path diversity (Section 13.10): Corrections on the slow path survive bursts on the fast path (different channel, different burst pattern). A burst on path A is covered by corrections from path B. $P(\text{both paths burst simultaneously}) = \epsilon_A \times \epsilon_B$ — negligible for independent paths.

No correction floor ($r_{\min}$) is needed. The global deficit and cross-path diversity provide burst protection without any per-path minimum. The taper self-corrects via BOCD if loss increases, and $P_{\text{lost}}(t)$ naturally produces Mehrotra's optimal policy (repair during uncertainty, retransmit after confirmation).

13.8 Interpolated Objective Function

One parameterized objective with weights from the protocol hint:

```
minimize:  $\hat{w}_{\text{lat}} \times \text{SUM}(x_i \times E_i) + \hat{w}_{\text{bw}} \times \text{SUM}(x_i \times r_i)$ 
           latency cost                bandwidth overhead cost

subject to:  $\text{SUM}(x_i) = 1$                 all source distributed
             $x_i \times \text{source\_rate} \leq B_{\text{eff}_i}$  per-path capacity
```

Realtime:	w _{lat} = 1.0, w _{bw} = 0.0	minimize latency at any bandwidth cost
Balanced:	w _{lat} = 0.5, w _{bw} = 0.5	balance latency and bandwidth
Bulk:	w _{lat} = 0.0, w _{bw} = 1.0	minimize bandwidth waste (overhead)

13.9 QoS Priority Cascade

When multiple protocol classes share the same paths, they pick in priority order from tightest to loosest latency requirement:

1. Realtime picks first: lowest E_i paths, up to its source volume
2. Balanced picks next: best remaining path capacity
3. Bulk gets the rest: whatever capacity remains

13.10 Cross-Path Retransmit

The shared retransmit buffer enables recovery across paths. When source symbol S_k was sent on path A and lost, any path's correction slot can retransmit it:

$$P(\text{retransmit on path } j) = P_{\text{lost}}(t_k, e_A)$$

Cross-path diversity: $P(\text{both fail}) = e_A \times e_j$. For $e_A=0.10$, $e_j=0.02$: $P(\text{both fail}) = 0.002$ — 50x improvement over single-path.

No explicit path routing needed. Cross-path retransmit emerges from the shared buffer. Each path independently generates correction symbols from its taper. When a correction slot produces a retransmit (via P_{lost}), it pulls from the shared buffer — which may contain symbols from any path. P_{lost} uses per-path ϵ , so symbols from lossier paths have higher P_{lost} and are retransmitted first.

Work-stealing analogy. Paths with spare Copa capacity (typically low-loss paths whose tapers need fewer corrections) have room to pull retransmits from the shared pool — like idle threads stealing work. High-loss paths are busy with their own corrections. This naturally routes retransmits through reliable paths without explicit selection.

Potential refinement: For latency-sensitive traffic, weighting retransmit pulls toward the lowest-RTT path would reduce recovery time. This follows the same interpolated objective as source scheduling (Section 13.8).

Appendix A: Summary of Key Formulas

Channel (Section 2):

$e = p/(p+q)$	average loss rate	[probability]
$B = 1/q$	mean burst length	[symbols]
$P(\text{burst} \geq t) = (1-q)^{t-1}$	burst survival	[probability]

Taper function (Section 4):

$\tau^*(t) = A^* \times (1-q)^t$	optimal taper function	[corrections/ symbol]
$A^* = r^* \times q$	taper amplitude ($p=100\%$)	[corrections/ symbol]

Burst variance correction (Section 8.3):

$\sigma^2_{\text{burst}} = 1 + 2(1-p-q)/(p+q)$	burst variance inflation	[dimensionless]
$\text{Var}_{\text{GE}}(K) = W \times e \times (1-e) \times \sigma^2_{\text{burst}}$	loss count variance	[symbols ²]

Optimal correction rate (Section 8.4):

Base: $r^* = e/(1-e) + z_{\text{delta}} \times \sqrt{e \times \sigma^2_{\text{burst}} / (W \times (1-e))}$	[ratio]
--	---------

With codec: replace e with e_{hat} (see Section 9.2)

$P_{\text{fec}} = \Phi(\sqrt{W} \times (r(1-e)-e) / \sqrt{e(1-e)(r+s_2\text{burst})})$ [probability]
 $z_{\text{delta}} = \text{normal_quantile}(1-\delta)$ [dimensionless]

Codec overhead (Section 9.2):

$e_{\text{codec_eff}} = e_{\text{codec}} \times (1-(1-e)^W)$ weighted codec overhead [probability]
 $e_{\text{hat}} = e + e_{\text{codec_eff}}$ effective loss rate [probability]

Three-variable optimization (Section 8.6):

Taper cutoff: $\tau(t) = 0$ for $t > T_{\text{cut}}$ ($\rho < 100\%$)
Correction rate with cutoff: $r = A \times (1-(1-q)^{\{T_{\text{cut}}+1\}}) / q$
Given $(\delta, \rho) \rightarrow r$: find T_{cut} from ρ , find A from δ , $r = A(1-(1-q)^{\{T_{\text{cut}}+1\}})/q$
Given $(r, \rho) \rightarrow \delta$: find T_{cut} from ρ , compute A from r , $\delta = e(1-P_{\text{fec}})P_{\text{arq}}/\rho$
Given $(r, \delta) \rightarrow \rho$: iterate T_{cut} until budget constraint met, $\rho = \text{recovery within } T_{\text{cut}}$

Per-symbol delivery (Section 6.3):

$P(\text{on-time}) = (1-e) + e \times P_{\text{fec}}$ [probability]
 $P(\text{late}) = e \times (1-P_{\text{fec}}) \times P_{\text{arq}}$ [probability]
 $P(\text{lost}) = e \times (1-P_{\text{fec}}) \times (1-P_{\text{arq}}) = 1-\rho$ [probability]
 $P_{\text{arq}} = 1 - (1-\rho) / (e \times (1-P_{\text{fec}}))$ [probability]

Recovery latency (Section 3.4):

$t_{\text{sym}} = \text{symbol_size} / \text{throughput}$ symbol transmission time [seconds]
 $t_{\text{fec}} = m / (A \times (1-e)) \times t_{\text{sym}}$ FEC recovery time [seconds]
 $t_{\text{recovery}_i} = P_{\text{fec}_i} \times t_{\text{fec}_i} + (1-P_{\text{fec}_i}) \times L_{\text{arq}_i}$ [seconds]
 $P_{\text{lost}}(t) = e / [e + (1-e) \times P(\text{RTT} > t)]$ loss confidence [probability]
 $P(\text{RTT} > t) = 1 - \Phi((t - \text{SRTT}) / \text{RTTVAR})$ [probability]
 $L_{\text{actual}} = \min(t_{\text{fec}}, \text{retransmit arrival})$ [seconds]

Retransmit buffer (Section 6.1):

$B_{\text{max}} = \text{ceil}(\ln(0.0001) / \ln(1-q))$ [symbols]
 $\text{buffer_max} = \text{source_rate} \times (\text{RTT} + B_{\text{max}}/(r \times (1-e)) \times t_{\text{sym}})$ [symbols,
 $\rho=100\%$
 $\text{buffer_max} = \text{source_rate} \times T_{\text{cut}}$ [symbols,
 $\rho < 100\%$

Per-slot decision (Section 5.4):

$P_{\text{retx}} = P_{\text{lost}}(t)$ (time-based
mixing)
with probability P_{retx} : send source retransmit (immediate
decode)
with probability $1-P_{\text{retx}}$: send repair symbol (FEC, any loss)
Optional refinement: $P_{\text{retx}} = P_{\text{lost}}(t) \times (1 - e_{\text{burst}})$
for long-burst scenarios (Appendix C.6)

Congestion control (Section 12):

Copa: $\text{rate} = 1 / (d_{\text{copa}} \times dq)$ [symbols/sec]
 $dq = \text{RTT}_{\text{current}} - \text{RTT}_{\text{min}}$ [seconds]
 $\text{source_rate} = \text{total_rate} / (1 + r^*)$ [symbols/sec]
 $\text{correction_rate} = \text{total_rate} \times r^* / (1 + r^*)$ [symbols/sec]

Multi-path scheduling (Section 13):

$E_i = \text{RTT}_i/2 + e_i \times t_{\text{recovery}_i}$ [seconds]
 $B_{\text{eff}_i} = C_i / (1 + r_i)$ [symbols/sec]
 $e_{\text{combined}} = \text{SUM}(C_i \times e_i) / \text{SUM}(C_i)$ [probability]
 $\text{deficit} = \text{SUM}_{\{\text{un-ACKed } s\}}(e_s)$ [expected
corrections]
cross-path: $r = e_{\text{source}} / (1 - e_{\text{correction}})$ [ratio]

$$\begin{aligned}
&\text{minimize: } w_{\text{lat}} \times \text{SUM}(x_i \times E_i) + w_{\text{bw}} \times \text{SUM}(x_i \times r_i) \\
&P(\text{cross-path retx}) = P_{\text{lost}}(t, e_{\text{src}}) \\
&P(\text{both paths fail}) = e_{\text{src}} \times e_{\text{retx}} \quad \quad \quad [\text{probability}]
\end{aligned}$$

Appendix B: Related Work

ACK-Based vs NACK-Based Recovery

Our model uses ACK-based (positive acknowledgment) feedback, the same approach that TCP has used successfully for over 40 years. In contrast, many real-time protocols (e.g., RTCP NACK [RFC4585]) use negative acknowledgment where the receiver explicitly reports losses.

Why ACK-based is more robust for unicast: - ACKs confirm what works; NACKs report what failed. If a NACK is lost, the sender never learns about the loss. If an ACK is lost, the sender simply retransmits after a timeout (self-healing). - SACK [RFC2018] provides the same information as NACKs (which symbols are missing) but derived from positive evidence (which symbols arrived). - The sender can infer losses from SACK gaps without requiring the receiver to detect and report them.

NACK-based approaches remain useful for multicast (one sender, many receivers) where ACK implosion is a concern [RFC4585]. For our unicast tunnel model, ACK+SACK is strictly superior.

Hybrid FEC-ARQ for Lossless Streaming

The closest prior work is Mehrotra, Li & Huang [Mehrotra2010], who solve the same core problem: minimize delivery delay for lossless, in-order streaming over a lossy channel using hybrid FEC+ARQ. Their key result: sometimes it is optimal to **preempt original data packets with FEC packets** — delaying new data to send proactive repair prevents an ARQ round-trip and reduces overall latency. This maps directly to our taper function concept.

Differences from our model: - They use a Markov Decision Process (MDP) formulation and compute the optimal policy numerically. We derive a closed-form taper function from the GE survival function, which is more directly implementable. - Their model doesn't account for burst correlation in the margin term (our σ^2_{burst} correction). - They don't address multi-protocol traffic classes or interleaving.

The precursor paper [Mehrotra2009] establishes the basic hybrid FEC-ARQ protocol framework.

Streaming Codes over Gilbert-Elliott Channels

Vajha et al. [Vajha2020] provide the first **analytical** (not simulation-only) bounds on block-erasure probability for streaming codes over GE channels. Previously, streaming codes were designed for a simplified proxy channel (the delay-constrained sliding window model from [Badr2017]) and then evaluated via simulation on GE. Their upper and lower bounds could be used to verify our P_{fec} predictions without simulation.

Very recent work [RLC_GE2025] analyzes Random Linear Codes (our RLC backend) specifically over GE channels, providing analytical performance characterization that directly applies to our system. A related result [Vajha2020b] formalizes the sliding window approximation of the GE channel used in streaming code design.

The foundational work on delay-constrained burst erasure correction [Martinian2004] established the theoretical framework that later streaming code constructions build upon. Krishnan & Ramkumar [Krishnan2020] provide the simplest rate-optimal streaming code construction (staggered diagonal embedding), demonstrating that optimal streaming codes need not be complex. Tambur [Rudow2023] is a production-quality implementation of streaming codes for videoconferencing, demonstrating 26% fewer decode failures and 35% less redundancy bandwidth — validating that streaming code theory translates to practical gains.

Tail Latency Optimization with Proactive FEC

CloudBurst [Zeng2021] uses proactive FEC over multipath to reduce p99 tail latency by 60-75% in commodity datacenters. Different setting (datacenter LAN vs our WAN/wireless focus) but the same core insight: proactive FEC-coded packets spread across paths, recovered from the first arrivals. They use rateless fountain codes; we use windowed RLC/streaming codes with a taper.

Joint FEC/ARQ under Gilbert-Elliott

Razavi et al. [Razavi2008] develop adaptive heuristic algorithms that jointly select FEC redundancy and ARQ persistence under GE channel conditions, applied to video streaming. They search the joint parameter space numerically. Our approach is more principled: the GE parameters directly determine the taper shape, and the tail latency constraint determines the amplitude analytically.

Water-Filling and Optimal Resource Allocation

The water-filling principle [Gallager1968] — allocating resources proportional to channel quality — is foundational in information theory. Our taper function applies this principle in the time domain: allocate correction density proportional to the conditional loss probability $(1-q)^t$. While water-filling is well-known for power allocation in OFDM, its application to correction symbol density matching the GE burst survival function appears novel.

What This Model Contributes

Aspect	Prior work	This model
Optimization target	Throughput, avg delay, or weighted cost	Tail latency only (= tail loss under 100% reliability)
FEC/ARQ balance	MDP or heuristic search	Closed-form from GE params + tail constraint
Taper function	Not formalized	GE survival function $\tau(t) = A(1-q)^t$
Burst correction	Not addressed	$\sigma^2_{\text{burst}} = 1+2(1-p-q)/(p+q)$
Protocol hint	Separate FEC/ARQ tuning knobs	Single knob: tail latency target δ
Correction rate formula	Numerical	$r^* = \epsilon/(1-\epsilon) + z_{\delta}\sqrt{(\epsilon \cdot \sigma^2_{\text{burst}}/(W(1-\epsilon)))}$
Feedback mechanism	NACK-based or separate FEC/ARQ	Unified correction symbols with ACK+SACK

Appendix C: Model Extensions from Related Work

The following extensions are motivated by concrete results from related research and represent improvements over the base model in Sections 1-11.

C.1 Correction Symbol Preemption of Source Data [Mehrotra2010] (resolved)

Mehrotra & Li show via MDP that during detected bursts, the optimal policy shifts from sending new source data to sending correction symbols. Our model produces the same behavior through two mechanisms:

1. **P_{lost}(t) during bursts:** Since the ACK hasn't had time to return (RTT hasn't elapsed), P_{lost} stays at the base rate ϵ — almost all correction slots are FEC repair. This is exactly Mehrotra's recommendation: send repair (not retransmit) during bursts.
2. **P_{lost}(t) after bursts:** Once ACKs reveal gaps (after RTT), P_{lost} rises to ~ 1 for lost symbols — correction slots switch to retransmit. Immediate decode at receiver, targeted recovery.
3. **Source preemption via r:** When BOCD detects increased ϵ , the taper amplitude A increases, and $\tau(t)$ naturally exceeds 1.0 at small t . This means more correction slots than source slots — source is effectively paused. No special $\tau > 1.0$ rule needed; it emerges from the formula.

The P_{lost} model and Mehrotra's MDP agree on the optimal policy. The difference is derivation: we arrive at the answer from a Bayesian posterior (P_{lost}) and rate adaptation (r^*), while Mehrotra uses dynamic programming.

C.2 Information Debt for Exact P_{fec} [RLC_GE2025]

The base model uses a normal approximation for P_{fec}. The information debt framework tracks the running deficit between received and needed symbols:

$$I_d(t) = \text{symbols_needed}(t) - \text{symbols_received}(t)$$

Recovery occurs when I_d returns to 0. The slot error probability is:

$$p_e = E\{\text{error_slots}\} / E\{\text{debt_cycle_length}\}$$

where error_slots counts time steps where $I_d \geq \zeta$ (the maximum recoverable debt, determined by window size and code rate).

Key finding: For systematic codes, successfully receiving a source symbol can sometimes **reduce** decodability of earlier lost symbols through Gaussian elimination interactions. Our normal approximation misses this effect.

Extension: Replace the normal-approximation P_{fec} with debt-tracking:

Current model: $P_{fec} \approx 1 - \Phi(-z)$ where z depends on (e, W, σ^2_{burst})

Extended model: $P_{fec} = 1 - p_e$ where p_e from debt Markov chain

The debt Markov chain has state space $\{0, 1, \dots, \zeta\}$ with transition probabilities depending on GE state and repair density. This is computable (finite-state Markov chain) though not as simple as the closed-form formula.

Status: The normal formula is used in the paper for analytical insight. For implementation, the exact $O(W^2)$ transfer matrix computation (see open point #4, resolved) provides the same precision as the debt model. Both are finite-state Markov chain computations over the GE channel — the debt model tracks decoder state, the transfer matrix tracks loss counts. Either gives exact P_{fec} for implementation use.

C.3 Analytical P_fec Bounds [Vajha2020]

Vajha et al. derive upper and lower bounds on block-erasure probability for streaming codes over GE without simulation. Their bounds depend on: - GE parameters (p, q) - Code rate $R = K/N$ - Decoding delay constraint T

Extension to verification (Section 11):

For our taper with rate r^* and window W :
Effective code rate $R = 1/(1+r^*)$
Delay constraint $T = W$

Vajha lower bound $\leq P(\text{erasure}) \leq$ Vajha upper bound

If our predicted P_{fec} falls within these bounds: ✓ model validated
If not: normal approximation is inadequate → use debt model (C.2)

This gives us an analytical verification path that complements simulation. Every verification method strengthens confidence in the model — simulation validates end-to-end behavior, Vajha bounds validate the P_{fec} formula analytically, and the exact $O(W^2)$ computation validates the normal approximation numerically.

C.4 Multipath Diversity Gain [Zeng2021]

CloudBurst sends FEC-coded packets across multiple paths and recovers from whichever path delivers first. This is a **diversity gain**: the same repair on two independent paths has survival probability $1 - \epsilon_1\epsilon_2$ instead of $1 - \epsilon_1$.

Extension: For multipath, the effective repair arrival rate is higher than single-path. With M independent paths of loss rates $\epsilon_1 \dots \epsilon_m$:

$P(\text{repair arrives on at least one path}) = 1 - \prod(e_i)$

Single path: $P(\text{arrive}) = 1 - e$
Dual path: $P(\text{arrive}) = 1 - e^2 \approx 1$ for small e

This means the multipath taper can use a **lower amplitude** A than single-path for the same P_{fec} target. The optimal multipath taper:

$\tau_{\text{multi}}(t) = A_{\text{multi}} \times (1-q)^t$

where $A_{\text{multi}} = A_{\text{single}} \times (1-e) / (1-\prod(e_i))$

For two paths with 5% loss each: $A_{\text{multi}} = A_{\text{single}} \times 0.95/0.9975 \approx 0.95 \times A_{\text{single}}$. Modest gain for similar paths, but significant when paths have different characteristics (one lossy WiFi, one reliable Ethernet).

Not applicable to our model. CloudBurst duplicates the same repair symbol across paths. Our model uses shared-buffer cross-path retransmit instead (Section 13.10): each path generates its own corrections, and the shared retransmit buffer enables cross-path recovery. The diversity gain $P(\text{both fail}) = \epsilon_A \times \epsilon_B$ is already captured in Section 13.10 without repair duplication.

C.5 DCSW Worst-Case Taper Floor

The Delay-Constrained Sliding Window model [Badr2017], [Fong2019] provides a **hard guarantee**: within any window of W symbols, at most B consecutive erasures or N random erasures can occur, and recovery must happen within delay T .

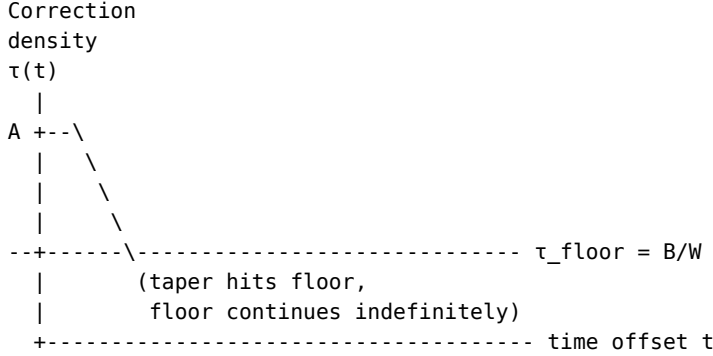
The streaming capacity for this model is $C(T,B) = T/(T+B)$ [Badr2017].

Extension: Regardless of the probabilistic taper, enforce a minimum correction density that survives the worst-case DCSW pattern:

$$\tau_{\text{floor}} = B / W \quad (\text{enough to survive one full burst per window})$$

$$\tau^*(t) = \max(A \times (1-q)^t, \tau_{\text{floor}}) \quad (\text{probabilistic taper with hard floor})$$

The floor ensures that even if the GE estimator underestimates burst length, we always have enough correction to survive at least B consecutive erasures.



Not applicable to our model. The DCSW floor is a hard guarantee from a worst-case adversarial channel model. Our model already handles burst protection through multiple mechanisms: (1) the taper's front-loaded density exceeds B/W for any reasonable r^* , (2) the retransmit buffer holds burst-lost symbols for ARQ recovery after detection, (3) cross-path diversity on multipath, and (4) backpressure guarantees delivery for $\rho=100\%$. Even if a burst overwhelms the taper entirely, ARQ kicks in — the retransmit buffer IS the hard guarantee. The DCSW floor is redundant.

Corrected total correction rate with floor (for reference):

$$r^* = \max(A/q, \tau_{\text{floor}} \times W) / W$$

In practice: A/q dominates when loss is high (large A).

Floor dominates when loss is low but bursts are long (large B , small e).

C.6 Burst-Aware Channel Discount (considered refinement)

The per-slot decision can optionally include a channel-state discount:

$$P_{\text{retx}} = P_{\text{lost}}(t) \times (1 - e_{\text{burst}})$$

where e_{burst} is a fast-reacting estimate of current channel loss (from GE state or fast EWMA). This suppresses retransmit during ongoing bursts, where repair is more flexible (covers any lost symbol).

Two separate loss estimates serve different purposes:

- ϵ (long-run, from BOCD): Bayesian prior for per-symbol loss confidence in P_{lost}
- ϵ_{burst} (fast, from GE state): current channel condition discount

This was considered for the primary model but deemed unnecessary because:

1. $P_{\text{lost}}(t)$ alone handles the primary FEC/ARQ regulation via timing
2. Before burst detection ($t < \text{RTT}$): P_{lost} is already low $\rightarrow$ mostly repair

3. After burst detection ($t > \text{RTT}$): burst is usually over $\rightarrow \epsilon_{\text{burst}}$ has dropped
4. The improvement is only significant for long bursts still ongoing at detection time ($\sim 1\text{ms}$ improvement)

The source/correction ratio was also considered for fast burst adjustment (two-speed taper: $A_{\text{effective}} = A_{\text{baseline}} \times (1 + \text{burst_boost})$), but the existing BOCD rate adaptation handles this with negligible delay.

Preserved here for future reference if extreme burst scenarios require it.

C.7 Summary of Extensions

Extension	From	Effect on correction rate	Effect on P_{fec} accuracy
Correction preemption	[Mehrotra2010]	Reduces delay during bursts	Improves burst recovery
Information debt	[RLC_GE2025]	More precise	Exact (Markov chain)
Analytical bounds	[Vajha2020]	Verification only	Bounds, not point estimate
Multipath diversity	[Zeng2021]	Reduces A_{multi}	Higher for same budget
DCSW taper floor	[Badr2017]	Hard minimum guarantee	Worst-case protection
Burst-aware discount	C.6	Modest improvement for long bursts	Suppresses retransmit during bursts

Appendix D: Open Questions

1. **Finite window truncation (resolved):** The taper is infinite-tailed but the encoder window has finite size W . After eviction, FEC repair is no longer possible — but the retransmit buffer (Section 5.3) still holds the exact source symbol for ARQ retransmission. The truncation error $(1-q)^W$ only affects the FEC component; the unified model's ARQ fallback covers it.
2. **Multi-path (resolved):** Each path has its own taper, Copa, and GE estimator (Section 13). Unified stream per path preserves interleaving. Scheduler adjusts per-path source/correction ratio (latency mode only). Global correction deficit tracks outstanding corrections. Cross-path retransmit via shared buffer. Burst protection during scheduler ratio adjustment is largely resolved: P_{lost} timing + BOCD adapts within $\sim 1\text{ms}$ (Section 13.7). Two-speed taper is an optional refinement for extreme scenarios (Appendix C.6).
3. **Interaction with congestion control (resolved):** Copa [Copa2018] controls total rate, taper controls source/correction split (Section 12). When $\text{source_rate} = \text{total_rate}/(1+r^*)$ drops below the app's minimum, the system signals back-pressure. Copa is preferred over BBR: no ProbeRTT, no FEC protection gaps, simpler formula, taper-compatible.
4. **Normal approximation validity (resolved).** The normal approximation uses the EXACT mean (We) and EXACT variance ($We(1-e)s_{\text{burst}}^2$) from the GE model — only the distribution SHAPE is approximated as a bell curve. For $W \gg B$ (all practical scenarios: $W=50$, $B=2-3$), the CLT makes this accurate. For edge cases (W close to B), the exact GE tail probability is computable via transfer matrix dynamic programming in $O(W^2)$ — trivially cheap (2500 operations for $W=50$). The normal formula provides analytical insight (clean, closed-form); the exact computation is recommended for implementation and validation.

5. **Optimal retransmit timing (resolved):** Replaced by the $P_{\text{lost}}(t)$ model (Section 3.4). No hard timeout — the repair/retransmit mix is determined probabilistically by $P_{\text{lost}}(t) = \epsilon / [\epsilon + (1-\epsilon) \times P(\text{RTT} > t)]$. This smoothly transitions from repair to retransmit as confidence in loss grows, naturally handles proactive retransmit at high loss rates, and minimizes expected waste at $E[\text{waste}] = P_{\text{lost}} \times (1 - P_{\text{lost}})$.

Remaining Open Points (full audit)

The following were identified by systematic review of the paper. Rated by difficulty: LOW = can be resolved with a sentence or formula, MEDIUM = needs a paragraph or derivation, HIGH = needs new analysis or algorithm design.

Core model gaps (critical):

6. **RTT distribution for P_{lost} not specified (resolved).** (Section 3.4) [LOW] P_{lost} uses $P(\text{RTT} > t)$ but never specifies the distribution. Assuming normal with SRTT and RTTVAR gives $P(\text{RTT} > t) = 1 - \Phi((t - \text{SRTT})/\text{RTTVAR})$. Needs one sentence.
7. **P_{arq} never defined (resolved).** (Section 6.3, 8.6, Appendix A) [MEDIUM] Derived from reliability target: $P_{\text{arq}} = 1 - (1 - \rho)/(e \times (1 - P_{\text{fec}}))$. See Section 6.3.
8. **P_{fec} : two contradictory models (resolved).** (Appendix E vs Section 8.2) [MEDIUM] Appendix E sections marked as preliminary (Poisson, superseded). Section 8.2 marked as canonical (Binomial/Normal). Progression preserved to show why per-symbol model fails.
9. **Poisson model error not characterized (resolved).** (Appendix E.4) [LOW] The transition from Poisson (Appendix E.3) to corrected (Section 8.2) says “too pessimistic” but doesn’t explain why. The error: Poisson treats each repair as independent, ignoring that repairs are deterministically generated by the taper. One paragraph.
10. **r^* formula: which version is canonical (resolved)?** (Section 8.4 vs 9.2) [LOW] Section 8.4 uses raw ϵ . Section 9.2 adds codec overhead ϵ_{codec} . The canonical formula should be r^* with $\epsilon_{\text{hat}} = \epsilon + \epsilon_{\text{codec}} \times P(\text{decoder})$. Needs a clarifying note in Section 8.4.
11. **T_{cut} computation algorithm (resolved).** (Section 8.6) [MEDIUM] Binary search on T_{cut} . $P(\text{recovered})$ is monotone in T_{cut} . Algorithm added to Mode 1. Converges in ~20 iterations.
12. **Mode 3 convergence (resolved).** (Section 8.6) [MEDIUM] Binary search on T_{cut} with monotonicity in ρ and r . Algorithm added to Mode 3. Convergence guaranteed.
13. **t_{recovery_i} undefined in multipath context (resolved).** (Section 13.5) [LOW] $E_i = \text{RTT}_i/2 + \epsilon_i \times t_{\text{recovery}_i}$ but t_{recovery_i} not defined for multipath. It should reference Section 3.4: $t_{\text{recovery}_i} = P_{\text{fec}_i} \times t_{\text{fec}_i} + (1 - P_{\text{fec}_i}) \times L_{\text{arq}_i}$. One sentence.

Implementation details (medium priority):

14. **Retransmit buffer saturation (resolved).** (Section 6.1) [LOW] Resolved: dual-mechanism model — T_{cut} age eviction + buffer_max size backpressure. Buffer is NOT bounded by encoder window. See Section 6.1.
15. **SACK timing and format (resolved).** (Section 6.2) [MEDIUM] Per-packet ACK (0.6% overhead). 5 fields: `cumulative_ack`, `sack_ranges` (cumulative within T_{cut}),

echo_timestamp, jitter_us (u32), cumulative_received. Gap pruning at T_cut. ACK loss self-healing via cumulative semantics.

16. **GE parameter initialization (resolved).** (Section 7.5) [LOW] Initial state: all counters = 0, start in Good state, use the Beta prior (weak uniform) until enough transitions observed. GE is_valid() already gates usage (existing code). One sentence.
17. **Copa parameter d tuning (resolved).** (Section 12.4) [MEDIUM] Renamed to d_copa to avoid confusion with tail latency delta. Default 0.5 sufficient. Adaptive d_copa (Copa+) is a negligible optimization (~1ms gain).
18. **Copa min_rtt refresh (resolved).** (Section 12.4) [LOW] Copa's natural oscillation refreshes min_rtt. Sliding window of 10s (same as BBR). If min_rtt seems stale (RTT consistently above by 2x), force a brief rate reduction. One paragraph.
19. **Back-pressure signaling (resolved).** (Section 12.7) [MEDIUM] Blocking write() like TCP — sender stops reading from source when buffer_max reached. Optional stats API for advanced applications. Implementation-specific, not part of core model.
20. **Scheduler burst protection floor (resolved).** (Section 13.7) [LOW] No floor needed. Global correction deficit + cross-path diversity handle burst protection when the scheduler reduces a path's correction ratio. The three options and floor discussion were removed.
21. **Cross-path retransmit path selection (resolved).** (Section 13.10) [LOW] No explicit path selection needed. Cross-path retransmit emerges from shared buffer + per-path P_lost. Paths with spare capacity naturally pull retransmits (work-stealing). Weighted path preference for latency-sensitive traffic noted as potential refinement.

Minor (notation, justification):

22. **z_d vs z_δ inconsistency (resolved).** (Section 8.4) [LOW] Line uses “z_d” and “d” where it should be “z_δ” and “δ”.
 23. **Beta decay 0.995 not justified (resolved).** (Section 7.3) [LOW] Standard value for slow-forgetting Bayesian update. Half-life ≈ 138 samples. Sensitivity is low — values 0.99-0.999 give similar results.
 24. **BOCD “5-15 batches” — what's a batch (resolved)?** (Section 4.5) [LOW] A batch = one ACK feedback cycle. With per-batch ACKs at ~10-100ms intervals, 5-15 batches = 50ms-1.5s adaptation time.
 25. **GE simplified model error bounds (resolved).** (Section 2.1) [MEDIUM] h_G=0, h_B=1 is exact for packet-level erasure (UDP checksums ensure no partial delivery). Not an approximation for our use case.
 26. **P_arq missing from Appendix A (resolved).** [LOW] Added P_arq formula to Appendix A.
-

Appendix E: Preliminary Poisson Model (superseded)

These sections were originally Section 6.2-6.5. They present a Poisson model that gives unrealistic results. The corrected Binomial/Normal model is in Section 8.2. This appendix is preserved for pedagogical value — it shows why the simpler per-symbol approach fails.

E.1 FEC Recovery Probability (preliminary — superseded by Section 8.2)

Note: Sections E.1-E.4 develop a preliminary Poisson model that gives unrealistic results. The model is superseded by the corrected Binomial model in Section 8.2. The preliminary derivation is kept to show why the simpler per-symbol approach fails and motivate the per-window model.

Consider a symbol lost at position 0. Correction symbols generated at offsets $t = 0, 1, 2, \dots$ each have: - Probability $\tau(t)$ of being generated (fractional: may or may not generate one) - Probability $(1-\epsilon)$ of surviving the channel

The expected number of correction symbols covering the lost position that arrive:

$$\begin{aligned} R(A, q) &= \sum_{t=0}^{W-1} \tau(t) \times (1-\epsilon) \\ &= A \times (1-\epsilon) \times \sum_{t=0}^{W-1} (1-q)^t \\ &= A \times (1-\epsilon) \times (1 - (1-q)^W) / q \end{aligned}$$

For large W (window much larger than burst length): $(1-q)^W \approx 0$, so:

$$R(A, q) \approx A \times (1-\epsilon) / q = r \times (1-\epsilon)$$

Recovery model: The number of useful correction symbols arriving is approximately Poisson(R). For FEC recovery, we need at least 1 repair symbol (plus codec overhead, see Section 9). Simplified to needing at least 1:

$$P_{\text{fec}}(A, q) = 1 - P(\text{Poisson}(R) = 0) = 1 - e^{-R}$$

E.2 Solving for A^* (preliminary)

Substituting into the constraint:

$$\begin{aligned} e \times (1 - P_{\text{fec}}) &\leq \delta \\ e \times e^{-R} &\leq \delta \\ e^{-R} &\leq \delta/e \\ -R &\leq \ln(\delta/e) \\ R &\geq \ln(e/\delta) \end{aligned} \quad (\text{note: } e > \delta, \text{ so } \ln(e/\delta) > 0)$$

Using $R \approx A \times (1-\epsilon) / q$:

$$A \times (1-\epsilon) / q \geq \ln(e/\delta)$$

$$A^* = q \times \ln(e/\delta) / (1-\epsilon)$$

$$r^* = A^*/q = \ln(e/\delta) / (1-\epsilon)$$

E.3 The Optimal Correction Rate Formula (preliminary — see Section 8.4)

$$r^* = \ln(e/\delta) / (1-e)$$

where:

e = average loss rate (from B0CD)

δ = tail latency target

r^* = optimal correction rate

Properties: - r^* depends only on ϵ and δ , not on q (burst length doesn't affect the TOTAL correction budget, only its distribution over time via the taper shape) - As $\delta \rightarrow 0$ (tighter tail): $r^* \rightarrow \infty$ (need infinite FEC for zero ARQ events) - As $\delta \rightarrow \epsilon$ (loose tail = every loss goes to ARQ): $r^* \rightarrow 0$ (no FEC needed) - As $\epsilon \rightarrow 0$ (perfect channel): $r^* \rightarrow 0$ (no correction needed)

The **taper amplitude** is:

$$A^* = r^* \times q = q \times \ln(e/\delta) / (1-e)$$

And the **complete taper function** is:

$$\tau^*(t) = A^* \times (1-q)^t = q \times \ln(e/\delta) / (1-e) \times (1-q)^t$$

E.4 Comparison with Information-Theoretic Minimum

The IT minimum (Shannon limit for the erasure channel [Shannon1948]) is:

$$r_{IT} = e / (1-e)$$

Why $\epsilon/(1-\epsilon)$, not just ϵ ? Correction symbols are also subject to channel loss. A correction sent to replace a lost source might itself be lost, needing another correction, which might also be lost:

$$e + e^2 + e^3 + \dots = e / (1-e)$$

At 30% loss: need 43% overhead (not 30%), because 30% of corrections are lost too. The $(1-\epsilon)$ denominator IS this geometric series. See also Section 13.4.

Our optimal rate:

$$r^* = \ln(e/\delta) / (1-e) = r_{IT} \times \ln(e/\delta) / e = r_{IT} \times \ln(1/\delta)/e + r_{IT} \times \ln(e)/e$$

The ratio $r^*/r_{IT} = \ln(\epsilon/\delta)/\epsilon$. This is the **unavoidable overhead** of targeting a tail latency of δ — it's the price of proactive protection.

For $\epsilon = 0.025$ (WiFi), $\delta = 1e-4$ (Realtime):

$$r^*/r_{IT} = \ln(0.025/0.0001) / 0.025 = \ln(250) / 0.025 = 5.52/0.025 = 221$$

This seems very high. Let's check: $r_{IT} = 0.025/0.975 = 0.0256$, so $r^* = 0.0256 \times 221 = 5.66$. That's 566% overhead — clearly too much.

The issue: The Poisson model (Appendix E.1-E.2) treats each repair symbol as an independent random event with probability $\tau(t)$ of existing. In reality, repair symbols are generated deterministically by the taper schedule — their positions are known, not random. The taper generates exactly r^*W repair symbols in a window of W , and each survives independently

with probability $(1-e)$. This is a Binomial process, not Poisson. The corrected model in Section 8.2 uses the Binomial/Normal approximation instead.

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